

Grey-Zone Sovereignty and Market Adaptation:

Financial Responses to Taiwan-Related Geopolitical Shocks

- *This paper DOES NOT argue that geopolitical risk does not matter. Instead, it argues that geopolitical risk is mediated by investor interpretation and adaptation.*
- *This paper evaluates whether observed market responses are consistent with the adaptation hypothesis.*

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Abstract

This research examines how financial markets respond to Taiwan-related geopolitical shocks under conditions of grey-zone sovereignty. Existing geopolitical-risk literature generally expects heightened geopolitical tensions to generate negative market outcomes by increasing uncertainty and perceived risk. However, market responses to major Taiwan-related events have not always followed this expectation.

Taiwan occupies a unique position within the international system. Whilst possessing many practical characteristics of sovereignty, it lacks broad diplomatic recognition and remains embedded within ongoing strategic competition between the United States and China. This condition of grey-zone sovereignty generates recurring geopolitical shocks that repeatedly expose financial markets to uncertainty.

Using an event-study approach, this research analyses market reactions to a series of Taiwan-related geopolitical events. The analysis focuses on abnormal returns in the Taiwan Stock Exchange (TWSE) and Taiwan Semiconductor Manufacturing Company (TSMC). Events are classified as either geopolitical-risk events or strategic-importance events in order to examine variation in market responses.

The findings indicate that geopolitical-risk events were associated with negative abnormal returns on average, whereas strategic-importance events were associated with positive abnormal returns on average. However, substantial variation existed across individual cases. Most notably, some events classified as geopolitical risks generated positive market reactions despite elevated objective tensions.

To explain this variation, this research develops a framework centred on grey-zone sovereignty, investor adaptation, and investor interpretation. The findings suggest that market responses are not determined solely by objective geopolitical risk. Instead, repeated exposure to recurring shocks may encourage investor learning and expectation formation, whilst investor interpretations of threat and opportunity influence the direction of market reactions.

This research contributes to the literature on geopolitical risk, financial markets, and Taiwan's international position by demonstrating that the relationship between geopolitical shocks and market outcomes is conditional rather than automatic. The findings suggest that recurring geopolitical tensions do not necessarily produce persistent financial damage when investors have adapted to similar events over time.

Intro

Geopolitical tensions are commonly expected to generate negative consequences for financial markets (Caldara & Iacoviello, 2022). Existing research on geopolitical risk generally argues that heightened uncertainty increases investor caution, reduces risk appetite, and contributes to negative market performance. Under this conventional expectation, periods of elevated geopolitical tension should be associated with increasingly negative market reactions.

Taiwan appears to represent a particularly important case for examining this relationship. Located at the centre of strategic competition between the United States and China, Taiwan occupies a unique position within the contemporary international system. Although Taiwan possesses many practical characteristics of sovereignty, it lacks broad diplomatic recognition and remains subject to competing political claims (Chiang, 2025; Shen, 2000). At the same time, Taiwan plays a critical role in the global economy due to its importance within international semiconductor supply chains (Bellocchi, 2023). As a result, Taiwan is simultaneously a geopolitical flashpoint and a strategically significant economic actor.

These conditions create what this research describes as grey-zone sovereignty: a persistent condition of political ambiguity that simultaneously generates recurring geopolitical risk and strategic economic importance. Unlike conventional interstate crises, Taiwan-related tensions frequently reappear through military exercises, diplomatic disputes, economic policies, and strategic competition. Consequently, financial markets are repeatedly exposed to similar geopolitical shocks.

Despite the apparent risks associated with these events, market reactions have not always followed conventional expectations. Some Taiwan-related geopolitical shocks generated negative abnormal returns, while others produced limited or even positive market responses. For example, the visit of United States House Speaker Nancy Pelosi to Taiwan in 2022 was associated with negative market reactions, whereas Joint Sword-2024B generated positive abnormal returns despite being classified as a geopolitical-risk event. Similarly, events associated with Taiwan's strategic importance within the semiconductor industry often produced positive market responses.

This variation presents an important puzzle. If geopolitical risk automatically generates negative market outcomes, why do some Taiwan-related geopolitical shocks fail to produce persistent financial damage? More specifically, why do some events associated with elevated objective risk generate neutral or positive market reactions?

This research argues that objective geopolitical risk alone cannot fully explain observed market responses. Instead, market reactions are mediated by investor interpretation and adaptation. Repeated exposure to Taiwan-related geopolitical events may encourage investor learning, expectation formation, and adaptation, reducing the severity of market responses over time. Furthermore, investors do not interpret all events in the same way. Some events are primarily perceived as threats, whereas others are viewed as opportunities linked to Taiwan's strategic economic importance. Consequently, market outcomes reflect not only objective risk but also how investors interpret and contextualise geopolitical developments.

To investigate this argument, this research employs an event-study approach examining market responses to a series of Taiwan-related geopolitical events. The analysis focuses on abnormal returns in the Taiwan Stock Exchange (TWSE) and Taiwan Semiconductor Manufacturing Company (TSMC). By comparing reactions across different categories of events, the research evaluates whether observed market behaviour is consistent with an adaptation-based explanation.

The findings suggest that geopolitical-risk events were associated with negative abnormal returns on average, whilst strategic-importance events were associated with positive abnormal returns on average. However, substantial variation existed across individual cases. These patterns indicate that the relationship between geopolitical shocks and market outcomes is conditional rather than automatic.

This research contributes to three strands of literature. First, it contributes to research on geopolitical risk by demonstrating that investor interpretation plays an important role in shaping financial outcomes. Secondly, it contributes to the research of financial responses to recurring geopolitical events by highlighting the potential importance of adaptation and expectation formation. Thirdly, it contributes to debates surrounding Taiwan's international position by conceptualising Taiwan as a case of grey-zone sovereignty and examining the financial implications of this condition.

The remainder of this dissertation proceeds as follows. Chapter Two reviews the relevant literature on geopolitical risk, sovereignty, geoeconomics, and market adaptation. Chapter Three

develops the theoretical framework. Chapter Four outlines the research design and methodology. Chapter Five presents the empirical findings. Chapter Six discusses the implications of the results and evaluates alternative explanations. Chapter Seven concludes.

Chapter 2. Literature Review

2.1 Sovereignty and Taiwan

The starting point for this dissertation is the problem of sovereignty. In classical political theory, sovereignty is associated with final authority over a political community. Hobbes located sovereignty in the capacity to establish order against insecurity, treating political authority as the necessary solution to the problem of disorder. In this tradition, sovereignty is tied to the ability to command, protect, and organise collective life. Weber's account shifts the emphasis towards the modern state as an institutional form, defined by the claim to the legitimate monopoly of physical force within a given territory. Taken together, Hobbes and Weber provide the conventional vocabulary through which sovereignty is often understood: authority, territory, coercive capacity, and political order.

Taiwan complicates this vocabulary. It possesses many of the practical attributes associated with sovereign authority. It has a democratically elected government, a professional military, independent economic policy, territorial administration, and functioning legal and bureaucratic institutions. In Weberian terms, it exercises many of the domestic functions of statehood. Yet the international status of Taiwan remains unresolved. It lacks broad diplomatic recognition, is claimed by the People's Republic of China, and is embedded within a strategic relationship in which the United States supports Taiwan without fully resolving the question of formal sovereignty. Taiwan therefore demonstrates that sovereignty cannot be understood only as domestic capacity. It is also a relational and international condition.

Wendt's constructivist approach is useful here because it treats sovereignty as a social institution rather than merely a material fact. Sovereignty depends not only on control, but also on recognition by others. States exist in a social world where status, legitimacy, and identity are produced through interaction. This insight is central to the Taiwan case. Taiwan's internal political authority is substantial, but its external recognition is limited. It exists as a polity whose practical autonomy is difficult to deny, but whose formal status is diplomatically constrained. Taiwan's position therefore highlights the gap between empirical sovereignty and recognised sovereignty.

Contemporary sovereignty scholarship sharpens this point. Krasner's account of sovereignty as organised hypocrisy is especially useful because it challenges the idea that formal sovereignty and political practice necessarily align. Sovereignty is often invoked as a stable legal principle, yet in practice it is repeatedly compromised, negotiated, violated, or selectively recognised. The importance of Krasner's argument for this dissertation is not that Taiwan should be treated as a legal exception, but that Taiwan exposes the divergence between formal sovereignty and political reality. Taiwan exercises many capacities associated with sovereign government, but its formal recognition remains constrained by diplomatic practice and

great-power politics. This divergence generates recurring politically significant events because actors continually signal, contest, avoid, or reinterpret Taiwan's status. Diplomatic visits, military exercises, recognition disputes, and policy statements become consequential precisely because the formal category of sovereignty does not resolve the political reality.

Taiwan sovereignty literature, including the work associated with Shen and Chiang, develops this problem more directly. Shen's contribution lies in connecting Taiwan's statehood question to the criteria and politics of international recognition. Taiwan cannot be understood simply through a checklist of state attributes, because its status is shaped by the political refusal of many states to recognise it formally. Chiang's more contemporary work extends this discussion into the present context of cross-strait relations, democratic identity, and Taiwan's changing international position. These scholars help explain why Taiwan occupies a distinctive position: it is not simply a normal state, but it is also not reducible to an object of Chinese policy. It is an autonomous political community whose sovereignty is practised domestically but contested internationally.

The limitation of this sovereignty literature is that it is primarily concerned with status, recognition, statehood, and identity. These are necessary foundations, but they do not directly explain financial-market behaviour. The literature explains why Taiwan's political position is ambiguous and why events involving Taiwan carry unusual symbolic and strategic weight. It does not explain why one sovereignty-related shock produces negative abnormal returns while another does not. Nor does it explain why investors may sometimes interpret Taiwan's contested position as a source of opportunity rather than only as a source of danger.

This dissertation therefore uses sovereignty theory not to resolve the legal status of Taiwan, but to develop the concept of grey-zone sovereignty. Grey-zone sovereignty refers to a persistent condition of political ambiguity that simultaneously generates recurring geopolitical risk and strategic economic importance. The concept moves beyond the question of whether Taiwan is or is not a state. Instead, it asks how Taiwan's ambiguous status generates repeated political events that markets must interpret. A diplomatic visit, a military exercise, an inauguration speech, or a recognition dispute becomes market-relevant because it occurs within an unresolved sovereignty environment. At the same time, Taiwan's status is tied to its economic centrality, especially in semiconductors. Grey-zone sovereignty therefore provides the political foundation for the dissertation's central puzzle: why do some Taiwan-related geopolitical shocks fail to generate persistent market damage despite elevated objective risk?

2.2 Geopolitical Risk and Financial Markets

The second body of literature concerns geopolitical risk and financial markets. Geopolitical risk research begins from the proposition that political conflict, military tension, terrorism, war, sanctions, and interstate rivalry can affect economic expectations and asset prices. Caldara and Iacoviello's work is central because it provides a systematic way to conceptualise and measure geopolitical risk. Their framework treats geopolitical risk as a form of uncertainty that can influence investment, consumption, volatility, and financial-market behaviour. The central intuition is that investors may discount assets more heavily when political

shocks raise uncertainty over future cash flows, supply chains, trade conditions, or macroeconomic stability.

Event-study literature extends this logic into an empirical strategy. If markets process new information quickly, returns around an event window can be used to assess how investors react to specific shocks. Political events such as elections, sanctions, terrorist attacks, military conflicts, or diplomatic crises can be examined by comparing actual returns with expected or benchmark-adjusted returns. This approach is particularly useful for the present dissertation because Taiwan-related geopolitical events have identifiable dates. The Pelosi Visit, Joint Sword exercises, export-control announcements, and TSMC policy events can be located in time, allowing the analysis to examine abnormal returns around those dates.

The geopolitical-risk literature gives strong reasons to expect Taiwan-related shocks to place downward pressure on markets. Taiwan is a site of persistent military pressure and diplomatic tension. Cross-strait escalation could disrupt trade, damage investor confidence, threaten supply chains, and raise the probability of conflict. It is therefore reasonable to expect higher geopolitical risk to be associated, on average, with more negative market reactions. This expectation is intuitive and is partly supported by the empirical findings. The Pelosi Visit, the China trade restrictions after the visit, the PLA exercises after the visit, Joint Sword 2023, and Strait Thunder-2025A all show patterns broadly consistent with the threat channel. Risk events are negative on average in the current event-study results.

However, this expectation should not be overstated. The geopolitical-risk literature does not require perfectly uniform negative outcomes in every event window, and sophisticated event-study research recognises that market responses depend on expectations, timing, benchmarks, co-occurring news, and the informational content of the event. The problem revealed by the Taiwan case is therefore more precise. The findings do not simply show occasional noise around an otherwise stable risk relationship. They show theoretically meaningful variation across events that are all connected to Taiwan's contested status. Some risk events are positive or mixed. Joint Sword-2024B is the most important example because it is a military demonstration but produces positive abnormal returns over the current event window. Lai's U.S. transit also fails to generate the negative pattern expected from a diplomatic-risk event. At the same time, BIS Advanced Semiconductor Export Controls, which is classified as a strategic-importance event, produces negative abnormal returns. These cases challenge a simple transmission model from geopolitical risk to market damage.

The limitation of the geopolitical-risk literature is therefore not that it is wrong, nor that it crudely assumes automatic market decline. Rather, the limitation is that it often leaves underdeveloped the interpretive process through which political risk becomes priced. Higher risk tends to increase uncertainty, and uncertainty often has negative financial effects, but the meaning of a specific event still has to be assessed by investors. A military exercise may be interpreted as a genuine escalation, but it may also be interpreted as a familiar form of grey-zone signalling. A technology-policy event may signal support for a strategic industry, but it may also signal restriction, fragmentation, or compliance costs. Event-study methods can detect

market reactions, but they do not by themselves explain why similar shocks produce different interpretations.

This dissertation builds on geopolitical-risk literature by retaining its core insight that political shocks matter for markets, while revising the expectation that risk is sufficient to predict the direction of market reaction. The Taiwan issue suggests that risk is not an automatic determinant of market outcome. It is a source of investor interpretation. This distinction is central to the grey-zone sovereignty framework. Taiwan-related events are risky because they emerge from unresolved sovereignty and strategic rivalry, but their market effects depend on whether investors interpret them as novel threats, repeated signals, opportunities, anticipated developments, or evidence of changing strategic value.

2.3 Geoeconomics and Semiconductor Politics

The third literature concerns geoeconomics, economic statecraft, and semiconductor politics. Baldwin's work on economic statecraft is important because it treats economic instruments as tools of political power. States use trade, finance, sanctions, investment restrictions, market access, and technology controls to influence other actors. In this view, economic policy is not separate from security. It is one of the means through which strategic competition is conducted. This insight is directly relevant to Taiwan, where economic interdependence, semiconductor production, export controls, and investment policy are inseparable from cross-strait and U.S.-China politics.

Wigell's geoeconomic framework similarly emphasises the fusion of economic and security logics. Geoeconomics refers to the use of economic instruments for strategic purposes and to the transformation of markets into arenas of power politics. The Taiwan case sits squarely within this literature. China's trade restrictions after the Pelosi Visit, U.S. semiconductor export controls, CHIPS Act subsidies, and TSMC's overseas expansion all show that economic events are not merely commercial. They are embedded in wider strategic competition. Taiwan's markets are therefore exposed not only to military risk, but also to financial statecraft, technology policy, and supply-chain politics.

Solis's work on economic security and regional order is relevant because it highlights the increasing importance of supply chains, technology governance, and economic resilience in contemporary international politics. In an era of strategic competition, states seek to secure critical technologies and reduce dependence on adversaries or vulnerable chokepoints. Taiwan is central to this shift because its semiconductor industry is both economically indispensable and geopolitically exposed. The location of advanced chip production in Taiwan creates a strategic dilemma: Taiwan's centrality makes it valuable, but that same centrality makes cross-strait risk globally consequential.

This economic-security literature is important because it explains why strategic importance can generate positive market expectations. Strategic industries are not valued only because they are profitable; they may also attract state support, subsidies, investment incentives, export-policy protection, infrastructure spending, and diplomatic attention. In the semiconductor sector, geopolitical rivalry can increase the perceived importance of firms and supply chains that

governments regard as critical. Investors may therefore interpret some Taiwan-related technology events as signals of industrial policy support, supply-chain protection, or long-run demand. Strategic importance can thus become a market-positive signal when it implies that states will mobilise resources to preserve, expand, or relocate critical capacity.

Bellocchi's work on the strategic importance of Taiwan helps connect this geoeconomic literature to the Taiwan-specific case. Taiwan's value does not derive only from its democratic identity or its position in U.S.-China rivalry. It also derives from its role in the technological infrastructure of the global economy. TSMC's position in advanced semiconductor manufacturing means that Taiwan is not just a possible site of conflict. It is also a critical node in the production of AI hardware, high-performance computing, and advanced industrial systems. Semiconductor politics therefore becomes a key part of the dissertation's explanation.

This literature explains why strategic-importance events can be associated with positive abnormal returns. TSMC Arizona expansion, TSMC CHIPS Act preliminary terms, and TSMC U.S. expansion are not ordinary corporate announcements. They signal industrial policy, supply-chain restructuring, and the strategic value of Taiwan-linked semiconductor capacity. Investors may interpret these events as evidence of policy support, long-run demand, subsidies, supply-chain protection, or the protection of strategic assets. This helps explain why strategic-importance events are positive on average in the current results.

Yet this literature also has limitations. It can overstate the positive implications of strategic importance if it assumes that being strategically valuable is always beneficial for markets. The BIS Advanced Semiconductor Export Controls case shows why that assumption is too simple. Export controls are clearly part of semiconductor geopolitics, but they can signal restriction and uncertainty as much as opportunity. Strategic importance may attract investment and policy support, but it may also produce regulatory burdens, supply-chain fragmentation, compliance costs, and exposure to great-power rivalry. In other words, semiconductor geopolitics does not automatically generate positive market reactions.

For the grey-zone sovereignty framework, the geoeconomic and semiconductor literatures provide the second force operating alongside risk. Taiwan's political ambiguity creates vulnerability, but its economic centrality creates strategic value. The dissertation's contribution is to show that both forces are present, and that neither determines market outcomes automatically. Strategic importance is a source of interpretation. Whether it becomes positive or negative depends on how investors read the event: as support, subsidy, constraint, opportunity, fragmentation, or evidence of future policy risk.

2.4 Investor Interpretation and Adaptation

The most important literature for the dissertation's mechanism concerns investor interpretation and adaptation. Whereas geopolitical-risk literature explains why risk should matter, and semiconductor politics explains why Taiwan can be strategically valuable, interpretation and adaptation explain why similar events can produce different reactions. Financial markets do not respond to objective events in a vacuum. They respond to events relative to expectations, prior information, recent precedent, and broader market narratives.

Taleb's work on uncertainty, rare events, and the limits of prediction is useful because it highlights the importance of surprise and non-linear reaction. Markets may be more vulnerable to events that fall outside expected patterns than to events that are severe but anticipated. The relevance to Taiwan is clear. A military exercise that occurs after repeated warnings and resembles previous exercises may be less informative than a sudden escalation that changes expectations. Severity and surprise are therefore analytically distinct. A high-risk event may generate a limited reaction if investors expected it, while a less severe event may produce a sharper response if it changes the perceived distribution of future outcomes.

Behavioural finance literature reinforces this point by showing that investors do not process information mechanically. Market reactions are shaped by attention, salience, prior beliefs, framing, representativeness, and learning. Investors may overreact to novel shocks and underreact to familiar ones. They may also use recent precedent to classify events. If earlier Taiwan Strait crises produced temporary disruption but not lasting market damage, later episodes may be interpreted as manageable unless they contain new information. This logic is central to the distinction between adaptation and anticipation. Anticipation implies that the event was expected and therefore priced before Day 0. Adaptation implies that repeated exposure changes the meaning investors assign to a class of events.

It is important, however, to distinguish adaptation from interpretation. Interpretation is the immediate mechanism that produces market reaction. Investors interpret an event as a threat, an opportunity, a familiar signal, a policy constraint, or a surprise, and those interpretations are reflected in prices. Adaptation is a deeper mechanism that changes interpretation over time. It occurs through repeated exposure, learning, expectation formation, and reduced novelty. A later military exercise may still be interpreted as risky, but if investors have seen similar exercises before and regard them as part of a recurring grey-zone repertoire, the threat interpretation may be weaker. Adaptation therefore influences interpretation; it is not a competing explanation to interpretation.

This clarification matters for the dissertation's theoretical argument. The sequence from the Pelosi Visit to Joint Sword 2023, Joint Sword-2024A, and Joint Sword-2024B suggests that repeated military demonstrations may become less damaging over time. The evidence is limited and should not be overstated, but the pattern motivates the adaptation hypothesis. Early shocks may be interpreted as more threatening because they are novel or less familiar. Later shocks may be interpreted as part of a repeated repertoire of grey-zone pressure. If investors learn that such events do not necessarily alter fundamentals, the interpretation of the event changes, and the market reaction may weaken.

At the same time, the adaptation literature has its own limitations. It is difficult to separate adaptation from anticipation. Later Joint Sword exercises may have produced weaker reactions not because investors learned to treat them as manageable, but because they were expected before they occurred. If a PRC military response after a Taiwan political event is widely anticipated, then the event window may miss the relevant repricing. This is why the dissertation

develops a surprise-coding protocol. Surprise must be measured explicitly if adaptation is to be distinguished from the simpler claim that markets had already priced the event.

Investor interpretation also helps explain why TWSE and TSMC sometimes diverge. The broad Taiwan equity market may be more sensitive to general geopolitical risk, while TSMC may be interpreted through semiconductor demand, AI expectations, strategic investment, and supply-chain policy. A single Taiwan-related event can therefore produce different asset-level reactions. This divergence is not a methodological inconvenience; it is theoretically important. It indicates that markets are not pricing a single undifferentiated Taiwan risk factor. They are interpreting different assets through different mixtures of threat, opportunity, adaptation, and surprise.

For the grey-zone sovereignty framework, the interpretation and adaptation literature provides the central mechanism. Grey-zone sovereignty produces recurring ambiguous events. Geopolitical risk and strategic importance provide the competing sources of meaning. Adaptation changes the way repeated events are understood over time, while interpretation is the immediate process through which those meanings are priced. Threat remains the baseline interpretation, opportunity explains some strategic-importance events, and surprise conditions the size of reactions. Adaptation, however, is the mechanism that most directly addresses the central puzzle because it explains why repeated high-risk events may become less damaging without assuming that objective risk has disappeared.

2.5 Literature Gap

The gap addressed by this dissertation is not that Taiwan has been ignored. Taiwan has been widely studied in relation to sovereignty, recognition, cross-strait relations, democratisation, deterrence, and U.S.-China rivalry. Nor is the gap simply that financial markets have ignored geopolitics. Geopolitical-risk research and event-study methods already examine how political shocks affect asset prices. The gap lies in how existing literature often leaves underdeveloped interpretive mechanisms linking geopolitical risk to market outcomes.

Much of the existing literature expects geopolitical risk to exert negative pressure on financial markets because risk increases uncertainty, raises the perceived probability of disruption, and can reduce investor confidence. This is a reasonable starting point, and the present findings partly support it. Yet the Taiwan issue reveals its limits. Risk events are negative on average, but not uniformly negative. Strategic-importance events are positive on average, but not uniformly positive. Some military events produce positive abnormal returns, while some technology events produce negative abnormal returns. These findings suggest that the relationship between political shock and market reaction is conditioned by interpretation.

Each literature reviewed above contributes to the dissertation, but each leaves part of the puzzle unresolved. Sovereignty literature explains why Taiwan's status is ambiguous, but not how investors price that ambiguity. Geopolitical-risk literature explains why risk should matter, but not why high-risk events sometimes fail to produce persistent damage. Geoeconomics explains why economic instruments and markets are part of strategic competition, but not why the direction of event-level market response varies. Semiconductor politics explains Taiwan's

strategic value, but not why strategic-importance events can be negative. Behavioural finance and adaptation literature explains expectation-based variation, but not why Taiwan produces recurring ambiguous shocks in the first place.

This dissertation contributes a grey-zone sovereignty framework linking sovereignty ambiguity, recurring geopolitical shocks, investor adaptation, investor interpretation, and market reactions. The contribution is not simply to add Taiwan as another case of geopolitical risk. It is to explain why Taiwan's unresolved sovereignty status generates a recurring stream of politically significant events, why repeated exposure may alter investor expectations, and why markets price those events through competing interpretations of threat, opportunity, adaptation, and surprise. The framework therefore connects the structural condition of sovereignty ambiguity to observable financial outcomes.

This contribution is distinct from the existing literature. Relative to sovereignty scholarship, the dissertation moves from recognition and status to market behaviour, showing how ambiguous sovereignty becomes economically consequential through recurring events. Relative to geopolitical-risk literature, it preserves the expectation that risk often has negative effects while specifying the interpretive mechanisms that condition those effects. Relative to geoeconomics and semiconductor politics, it shows that strategic importance is not merely a background feature of Taiwan's economy but a market-relevant signal that can generate both positive expectations and new constraints. The novelty lies in integrating these claims into a single framework rather than treating sovereignty, risk, economic statecraft, and investor behaviour as separate domains.

The contribution to International Relations is to connect sovereignty ambiguity and grey-zone pressure to observable financial-market behaviour. Rather than treating sovereignty as only a legal or diplomatic question, the dissertation asks how ambiguous sovereignty becomes priced through market reactions. The contribution to International Political Economy is to show how security, finance, economic statecraft, and semiconductor supply chains interact in a single case. Taiwan is not merely a geopolitical risk site. It is also a strategic economic node. Markets appear to respond to both dimensions.

The dissertation therefore advances a more conditional account of geopolitical risk. It does not reject the proposition that risk damages markets. Instead, it argues that this relationship is mediated by investor interpretation and shaped over time by adaptation. Repeated shocks may reduce novelty. Strategic-importance events may produce opportunity or constraint. Surprise may alter the magnitude of reaction. This framework is particularly suited to Taiwan, but it also has broader implications for other cases where political ambiguity, strategic competition, and economic centrality coexist.

Chapter 3: Theoretical Framework

3.1 Grey-Zone Sovereignty as the Independent Variable

This chapter develops the theoretical framework that guides the dissertation. The central claim is that Taiwan's grey-zone sovereignty operates as the independent variable because it is the underlying political condition from which the observed geopolitical events emerge.

Grey-zone sovereignty is defined here as a persistent condition of political ambiguity that simultaneously generates recurring geopolitical risk and strategic economic importance. The concept is not used simply as a descriptive label for Taiwan's unusual international status, nor as a legal argument about whether Taiwan should be classified as a state. It is used as an explanatory framework for understanding why Taiwan-related events repeatedly become financially salient.

Taiwan possesses many attributes normally associated with sovereign statehood: democratic institutions, independent policy-making capacity, territorial administration, a military, and a highly developed economy. Yet Taiwan lacks broad formal diplomatic recognition and remains subject to the People's Republic of China's sovereignty claim. The United States supports Taiwan's security without resolving the question of formal statehood. This produces a political condition that is neither ordinary sovereignty nor simple dependence. Taiwan exercises substantial autonomy, but that autonomy is persistently contested.

Grey-zone sovereignty should therefore be treated as the independent variable rather than as background context. It precedes the specific events analysed in the event study and explains why those events occur in the first place. A Pelosi visit, a Joint Sword exercise, an inauguration speech, an export-control announcement, or a TSMC investment decision becomes market-relevant because it takes place within a persistent structure of unresolved sovereignty, U.S.-China competition, and Taiwan's strategic economic centrality. Grey-zone sovereignty is the upstream condition that makes a class of recurring Taiwan-related shocks possible.

The dependent variable is financial-market reaction, measured through abnormal returns for the TWSE and TSMC around Taiwan-related event windows. The intervening process is investor interpretation. Grey-zone sovereignty does not directly determine returns in a mechanical way. Rather, it generates events that investors interpret as threat, strategic importance, or familiar repetition. The framework therefore moves from political condition to recurring event, from recurring event to investor interpretation, and from investor interpretation to market reaction.

This ordering matters because it prevents the analysis from reducing Taiwan to a simple risk variable. The findings already suggest that such a model is insufficient. Risk events are negative on average, but not uniformly negative. Strategic-importance events are positive on average, but not uniformly positive. Grey-zone sovereignty is therefore best understood as a generator of ambiguous market signals whose effects depend on interpretation.

3.2 From Grey-Zone Sovereignty to Recurring Shocks

Grey-zone sovereignty generates recurring shocks because Taiwan's political ambiguity is not episodic. It is a durable condition of international politics. Taiwan's unresolved status creates incentives for repeated signalling by

Figure 3.1. Theoretical Framework

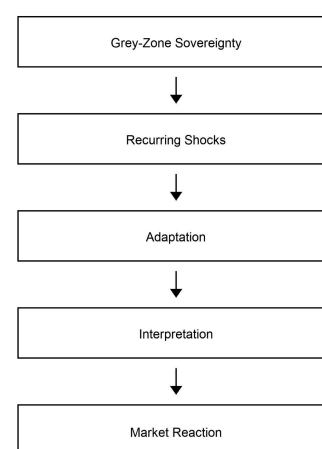


Figure 3.1. Theoretical Framework.

Note: The framework proposes that Taiwan's grey-zone sovereignty generates recurring geopolitical shocks. Repeated exposure may encourage adaptation, which influences investor interpretation and ultimately shapes market reactions.

multiple actors. The People's Republic of China has incentives to demonstrate that Taiwan remains within its claimed sovereignty domain and to deter moves that could be interpreted as formal independence. Taiwan's government has incentives to maintain autonomy, deepen international partnerships, and signal democratic legitimacy. The United States has incentives to support Taiwan while avoiding an explicit transformation of Taiwan's legal status. Other states and firms have incentives to engage Taiwan economically while managing diplomatic and strategic risk. These overlapping incentives produce recurring events rather than isolated crises.

The recurrence of Taiwan-related shocks is built into the structure of the issue. Military demonstrations, diplomatic protests, sanctions, export controls, political speeches, and semiconductor-policy announcements are not random disturbances. They are recurring expressions of the unresolved sovereignty environment. Each event may have a discrete date suitable for event-study analysis, but the class of events is generated by a continuing political condition. In a one-off crisis, markets might react primarily to severity. In a recurring grey-zone environment, markets may also react to whether an event resembles earlier episodes and whether it changes expectations.

This helps connect geopolitical risk with strategic importance. Taiwan's grey-zone sovereignty creates risk because its status remains contested and cross-strait escalation would have severe consequences. At the same time, the same ambiguity reinforces Taiwan's strategic importance. Taiwan is not merely exposed to U.S.-China competition; it is one of the key sites through which that competition is organised. Its semiconductor sector, especially TSMC, gives Taiwan global economic significance. Events connected to Taiwan may therefore signal not only danger but also protection, policy support, or technological centrality.

This dual character is central to the framework. A standard geopolitical-risk account would classify Taiwan-related military and diplomatic events as risk shocks and expect negative market reactions. This dissertation accepts that expectation as a baseline but argues that it is incomplete. Some Taiwan-related events heighten the perceived probability of coercion or conflict. Others demonstrate that Taiwan remains strategically indispensable. Some do both.

Recurring shocks also create the conditions for adaptation. If markets observe repeated crisis patterns, military exercises, diplomatic protests, and semiconductor-policy interventions, investors may begin to distinguish between shocks that materially alter fundamentals and shocks that belong to a familiar repertoire of grey-zone signalling. This does not mean that objective risk disappears. It means that market interpretation may change as similar events recur.

3.3 Adaptation and the Changing Interpretation of Risk

Adaptation is the primary mechanism in the theoretical framework because it addresses the dissertation's central puzzle most directly. The puzzle is not whether Taiwan-related risk events ever generate negative returns. They do. The puzzle is why some high-risk events fail to generate persistent market damage. Adaptation offers a parsimonious explanation: repeated grey-zone shocks become less novel, and less novel shocks are less likely to be interpreted as fundamental changes in the risk environment.

In this framework, adaptation refers to a change in investor interpretation produced by repeated exposure to similar events. Early or unusual events are more likely to be interpreted as threats because investors have less precedent for evaluating their implications. The Pelosi Visit and the immediate PLA response in 2022 can be understood in this way. Later exercises, such as Joint Sword 2023, Joint Sword-2024A, and Joint Sword-2024B, occurred within a more established pattern of Chinese military signalling around Taiwan-related political events. As this pattern became more familiar, investors may have become more capable of distinguishing between symbolic coercive signalling and a material change in the probability of conflict.

Adaptation does not imply complacency. It does not mean investors believe that cross-strait risk is unimportant. Rather, it means that repeated exposure can reduce the informational shock value of an event. A military exercise may still be objectively serious, but if it resembles previous exercises and does not appear to alter expected fundamentals, markets may react less negatively than a simple risk model would predict. This distinction separates objective risk from perceived market novelty. The same event can remain strategically serious while becoming less surprising or less informative to investors.

Adaptation also differs from opportunity and anticipation. Opportunity refers to positive interpretation based on strategic economic value, policy support, or supply-chain importance. Adaptation refers to the weakening of negative interpretation as events become familiar. Anticipation means that an event was expected before the event window and may already have been priced into markets. These mechanisms can produce similar empirical patterns. If Joint Sword-2024B generated a positive or muted abnormal return, this may indicate adaptation, but it may also indicate prior pricing, favourable semiconductor conditions, or benchmark limitations.

3.4 Investor Interpretation and Market Reaction

The theoretical framework places investor interpretation between geopolitical events and financial-market reaction. This is the central departure from a mechanical risk model. Markets do not respond to geopolitical events as objective facts alone. They respond to the expected implications of those events for future cash flows, discount rates, policy support, supply-chain stability, sector performance, and conflict probability. A market reaction is therefore an interpreted response to information.

Investor interpretation can take at least four forms in this dissertation. The first is threat: the perception that an event raises the probability of conflict, coercion, disruption, or policy instability. Threat should generally be associated with negative abnormal returns, particularly for Taiwan-wide market indicators such as the TWSE. The second is opportunity: the perception that an event signals Taiwan's strategic value, policy support for semiconductor capacity, supply-chain resilience, or long-term demand for technology firms. Opportunity should generally be associated with positive abnormal returns, particularly for TSMC.

The third form is adaptation: the interpretation of an event as part of an established grey-zone repertoire rather than as a new escalation. Adaptation should reduce the negative

effect of risk events over time, especially within repeated event families such as PLA military demonstrations after sovereignty-related political triggers. The fourth form is surprise: whether an event changes expectations. A severe event that was fully anticipated may produce little immediate abnormal return, while a less severe but unexpected event may produce a larger reaction. Surprise therefore affects the magnitude of market movement, while threat and opportunity more directly shape its direction.

This interpretive framework explains why the same event category may generate different outcomes. Military demonstrations are expected to be negative under a threat model, but they may become less negative if they are repeated, anticipated, or interpreted as containing signalling. Semiconductor-policy events are expected to be positive under an opportunity model, but they may be negative if investors interpret them as restrictive, costly, or evidence of technological fragmentation. The BIS Advanced Semiconductor Export Controls case is important because it belongs to semiconductor geopolitics, yet its market meaning may be closer to constraint than opportunity.

Interpretation also explains why TWSE and TSMC may respond differently. The TWSE captures broader Taiwan market exposure and may be more sensitive to general country risk. TSMC captures firm-level and sector-specific expectations about semiconductor demand, supply-chain restructuring, industrial policy, and AI-related growth. A Taiwan-related event can therefore be negative for the broader market while less negative or even positive for TSMC, or vice versa. This asset-specific variation supports the argument that markets are not pricing a single undifferentiated Taiwan risk factor.

3.5 Adaptation as an Inferred Mechanism

Adaptation is not directly observed in the data. The dissertation observes event classifications, abnormal returns, cumulative abnormal returns, event repetition, and preliminary surprise coding. It does not observe investor beliefs directly. This is why adaptation must be treated as an inferred mechanism rather than a demonstrated psychological process. The empirical evidence can show that later repeated shocks generate weaker negative abnormal returns than earlier shocks. It can show that a high-risk event such as Joint Sword-2024B does not behave as a simple risk model would expect. It can show that patterns are consistent with changing interpretation. It cannot, by itself, prove that investors consciously adapted.

This distinction is essential for the credibility of the argument. A dissertation-level theory should not treat market outcomes as transparent evidence of investor psychology. Abnormal returns are observable; interpretation is inferred. The inference is strengthened when events belong to a recognisable repeated family, later events are less negative than earlier comparable events, the pattern survives benchmark adjustment, rival explanations are considered, and surprise coding helps distinguish adaptation from prior pricing.

The current framework therefore presents adaptation as a plausible and testable interpretation of observed market patterns. It does not treat adaptation as proven. The language of the theory is deliberately conditional: repeated grey-zone shocks may become less damaging

because investors may learn to interpret them as familiar and contained. The theory organises the evidence and generates hypotheses for further testing, but it remains open to rival explanations.

3.6 Difference from Standard Geopolitical-Risk Theory

The framework differs from standard geopolitical-risk theory in three ways. First, it treats geopolitical risk as mediated rather than direct. Standard geopolitical-risk theory often implies that greater risk should generate more negative market outcomes because uncertainty raises discount rates, lowers expected investment, and increases the probability of disruption. This dissertation accepts that logic as a baseline but argues that it does not travel cleanly to a recurring grey-zone issue such as Taiwan. Risk must be interpreted before it becomes a market reaction.

Second, the framework treats Taiwan's political ambiguity as dual rather than purely negative. Grey-zone sovereignty generates vulnerability, but it also generates strategic importance. Taiwan is exposed because its status is contested; it is valuable because its semiconductor centrality makes it indispensable to U.S.-China competition and global technology supply chains. Standard risk theory captures the vulnerability side more easily than the strategic-importance side.

Third, the framework introduces repetition and adaptation. Standard geopolitical-risk theory is strongest when analysing discrete shocks that introduce new uncertainty. Taiwan-related events, however, often repeat. Repetition changes the informational content of a shock. A repeated exercise or diplomatic protest may be risky in an objective sense but less informative than an unprecedented escalation. This is why the dissertation gives adaptation a central role. It explains why the relationship between objective risk and market damage may weaken over time.

The resulting theory can be summarised as follows. Taiwan's grey-zone sovereignty produces recurring geopolitical and geoeconomic shocks. These shocks contain ambiguous market signals. Investors interpret them through threat, opportunity, adaptation, and surprise. Market reactions depend on this interpretation rather than on objective event type alone. The framework therefore predicts heterogeneity: risk events should be negative on average, but not uniformly negative; strategic-importance events should be positive on average, but not uniformly positive; repeated shocks should become less damaging when they resemble prior grey-zone patterns and do not change expected fundamentals.

3.7 Hypotheses

The framework generates four hypotheses for empirical evaluation. These hypotheses are not claims that the dissertation has already proven. They state the expectations implied by the theory and provide a structure for interpreting the event-study results.

H1, the threat hypothesis, is that Taiwan-related geopolitical-risk events are associated with negative abnormal returns. Events involving military demonstrations, diplomatic shocks, economic coercion, or sovereignty-related political signalling should, on average, produce more

negative TWSE and TSMC abnormal returns than events not interpreted as threats.

H2, the strategic-importance hypothesis, is that Taiwan-related strategic-importance events are associated with more positive abnormal returns. Events involving semiconductor geopolitics, CHIPS Act support, TSMC strategic investment, supply-chain restructuring, or technology competition should, on average, produce more positive abnormal returns, especially for TSMC.

H3, the adaptation hypothesis, is that repeated Taiwan-related grey-zone shocks are associated with weaker negative abnormal returns over time. As event repetition increases within comparable event families, especially military demonstrations and sovereignty-related signalling, cumulative abnormal returns should become less negative than a simple geopolitical-risk model would predict.

H4, the surprise hypothesis, is that more surprising Taiwan-related events are associated with larger abnormal-return movements. Surprise should affect the magnitude of reaction, whether negative or positive, because events that change expectations should contain more new information than events that were fully anticipated.

These hypotheses define the dissertation's theoretical claim. Grey-zone sovereignty is the structural independent variable because it generates the recurring ambiguous events that markets must price. Adaptation is the primary mechanism because it explains why repeated high-risk events may become less damaging. Investor interpretation is the intervening process because it determines whether a given event is priced as threat, opportunity, familiar repetition, or surprise.

Chapter 4. Research Design

4.1 Research Strategy

This dissertation uses an exploratory event-study research design to examine how Taiwan-related geopolitical shocks are associated with financial-market reactions. The research question asks how Taiwan-related geopolitical events affect markets through competing investor interpretations of threat, strategic importance, surprise, and adaptation. The purpose is not to prove that investors adapt to geopolitical risk. Rather, the

Figure 3.1. Conceptual Framework

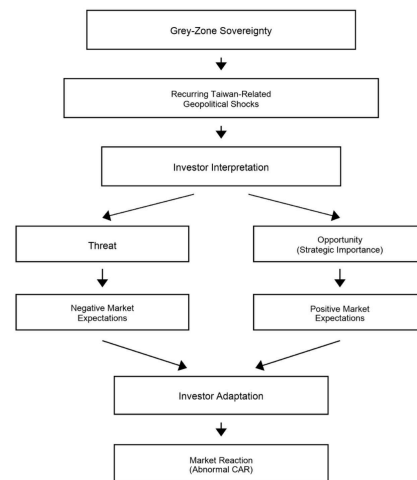
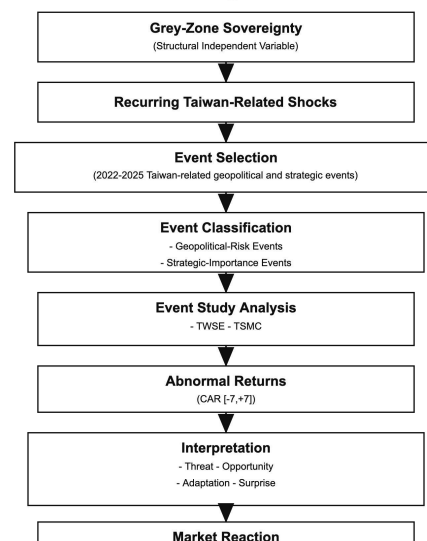


Figure 3.1. Conceptual Framework.

Note: The framework proposes that grey-zone sovereignty generates recurring geopolitical shocks, which investors interpret through threat and opportunity mechanisms. Repeated exposure may contribute to investor adaptation, influencing abnormal market reactions.

Research Design Framework



objective is to evaluate whether observed market responses are consistent with an adaptation-based interpretation while taking rival explanations seriously.

The design follows from the theoretical framework developed in Chapter 3. Grey-zone sovereignty is treated as the upstream political condition that generates recurring Taiwan-related shocks. These shocks may be interpreted by investors as threats, as signs of strategic importance, as familiar repeated signals, or as surprises that change expectations. The empirical task is therefore to observe market reactions around identifiable Taiwan-related events and assess whether the direction and magnitude of abnormal returns align with the theoretical expectations. The event study is appropriate because the project is concerned with dated political events and short-run financial responses.

The analysis is descriptive and interpretive rather than causal in a strong econometric sense. Taiwan-related events rarely occur in isolation, and financial markets are affected by many simultaneous forces. For this reason, the dissertation does not claim that each event caused a particular return. Instead, it asks whether the pattern of market responses is more consistent with a simple geopolitical-risk model or with a more conditional model in which investor interpretation and adaptation mediate the effect of political shocks. This distinction is important. The empirical strategy can identify patterns that support, complicate, or weaken the theory. It cannot directly observe investor beliefs.

The research design combines three elements. First, it constructs a transparent event sample of Taiwan-related geopolitical and strategic-importance events. Second, it classifies events according to mechanism and category. Third, it calculates raw and benchmark-adjusted returns around event windows. The resulting evidence is used to compare risk events with strategic-importance events, to identify deviant cases, and to examine whether repeated grey-zone shocks appear to become less damaging over time.

4.2 Taiwan as a Critical Case

Taiwan is a critical case because it brings together sovereignty ambiguity, geopolitical risk, economic statecraft, and semiconductor centrality in unusually concentrated form. It possesses many practical attributes of sovereignty, including democratic institutions, independent policy capacity, territorial administration, and military capability. Yet its formal international recognition remains limited, and its status is continually contested by the People's Republic of China. This creates the condition identified in this dissertation as grey-zone sovereignty: a persistent political ambiguity that simultaneously generates recurring geopolitical risk and strategic economic importance.

Taiwan is critical for geopolitical-risk theory because conventional expectations would predict that cross-strait tension, military pressure, diplomatic confrontation, and coercive economic measures should place downward pressure on financial markets. Taiwan is not a peripheral security issue. It is one of the most visible flashpoints in U.S.-China relations. A serious escalation could disrupt trade, damage investor confidence, threaten supply chains, and

create major regional-security consequences. If geopolitical risk should produce negative financial reactions anywhere, Taiwan is a plausible case in which to observe that relationship.

At the same time, Taiwan is critical for IPE and geoeconomics because it is not only a site of risk. It is also a strategic economic node. TSMC and Taiwan's semiconductor ecosystem sit at the centre of advanced chip manufacturing, AI infrastructure, and global technology competition. This means Taiwan-related events may communicate more than danger. They may also signal policy support, supply-chain protection, industrial subsidies, or the long-run value of strategic technologies. Taiwan therefore allows the dissertation to examine the tension between risk and strategic importance within the same case.

The case is especially useful for studying adaptation because Taiwan-related shocks recur. Military demonstrations, diplomatic protests, election-related warnings, sovereignty statements, U.S.-Taiwan interactions, export controls, and semiconductor-policy announcements are not one-off events. They are repeated manifestations of an unresolved sovereignty condition. This recurrence makes it possible to ask whether later shocks are interpreted differently from earlier shocks. Taiwan therefore functions as a critical case for testing whether markets respond only to objective risk, or whether repeated exposure changes the way risk is interpreted.

4.3 Event Selection

The event sample was constructed to include Taiwan-related events between 2022 and 2025 that could plausibly affect investor perceptions of Taiwan-related geopolitical risk or strategic economic importance. The inclusion criteria required that an event involve at least one of the following: Taiwan Strait military activity, Taiwan diplomatic relations, Taiwan-U.S. relations, China-Taiwan economic coercion, semiconductor geopolitics, export controls, AI supply-chain competition, or cross-strait crisis dynamics. The event also had to have a clearly identifiable date suitable for event-study analysis.

The first stage produced a broader candidate universe of Taiwan-related events. This included military exercises, diplomatic shocks, political signalling, economic coercion, export-control decisions, and strategic semiconductor events. Events were then reviewed for relevance to the research question. Routine business announcements, conferences, general China policy actions without a direct Taiwan connection, and corporate investment announcements without a clear geopolitical or strategic-importance component were excluded from the event-study sample. This exclusion rule is important because the dissertation studies market responses to politically meaningful shocks, not all news involving Taiwan or technology firms.

The selection process also addressed event-window contamination. Taiwan-related events often cluster. For example, the Pelosi crisis included prior warnings, the visit itself, trade restrictions, and PLA exercises within a narrow period. Treating each item as fully independent would overstate the amount of information in the sample and risk double-counting the same crisis episode. The cleaner sample therefore prioritises the event most likely to generate direct market repricing within severe overlap clusters. In the Pelosi cluster, the Pelosi Visit is treated as the preferred repricing event. In the Tsai-McCarthy and Joint Sword 2023 cluster, Joint Sword

2023 is preferred. In the Lai inauguration and Joint Sword-2024A cluster, Joint Sword-2024A is preferred. In the Lai National Day speech and Joint Sword-2024B cluster, Joint Sword-2024B is preferred.

The project therefore distinguishes between the broader calculated event set and the overlap-reduced event-study sample. The broader set is useful for descriptive exploration and for identifying patterns across more observations. The overlap-reduced sample is more appropriate for cautious interpretation because it reduces contamination across overlapping windows. This distinction reflects a central trade-off in the dissertation: increasing the number of events improves descriptive coverage, but reducing overlap improves interpretability.

Events with incomplete market data were excluded from estimation when the required event-day or event-window observations were unavailable. The final overlap-aware sample includes risk events and strategic-importance events, with risk events more numerous than strategic-importance events. This imbalance is substantively meaningful because Taiwan's grey-zone sovereignty produces more frequent military and diplomatic shocks than major semiconductor-policy shocks. It is also a limitation because it makes comparisons between mechanisms sensitive to a small number of strategic-importance observations.

4.4 Event Classification

Events are classified in two main ways: by event category and by mechanism. Event categories identify the substantive form of the event, such as Military Demonstration, Diplomatic Shock, Political Signalling, Economic Coercion, or Technology Competition. Mechanism identifies the theoretical interpretation most relevant to the event: Risk or Strategic Importance.

Risk events are those that plausibly increase perceived geopolitical uncertainty around Taiwan. Military demonstrations are classified as risk events because they involve coercive signalling, operational pressure, or visible escalation in the Taiwan Strait. Diplomatic shocks are classified as risk events when they raise cross-strait tension, trigger PRC warnings, or challenge Taiwan's contested international status. Political-signalling events are classified as risk events when they involve sovereignty claims, independence warnings, or official statements likely to affect perceptions of conflict probability. Economic-coercion events are classified as risk events when they use trade or economic measures to punish or pressure Taiwan.

Strategic-importance events are those that highlight Taiwan's role in critical technologies, semiconductor supply chains, or U.S.-China technology competition. TSMC-related investment announcements, CHIPS Act developments, and major semiconductor export controls are classified as strategic-importance events because they signal Taiwan's central position in global technology politics. These events may generate positive market expectations if investors interpret them as evidence of state support, subsidies, industrial policy protection, or long-run demand for advanced chips. However, this classification does not assume a positive market reaction. BIS Advanced Semiconductor Export Controls are treated as a strategic-importance event because they concern semiconductor geopolitics, but they may be interpreted negatively if investors view them as restrictive, fragmenting, or costly.

The classification scheme therefore separates event type from expected outcome. Risk events are expected to be negative on average because threat, coercion, and escalation tend to increase uncertainty. Strategic-importance events are expected to be positive on average because strategic industries may attract investment, policy support, and supply-chain protection. Yet both expectations are probabilistic rather than deterministic. A risk event may generate muted or positive returns if investors interpret it as familiar, anticipated, or contained. A strategic-importance event may generate negative returns if it signals regulation, fragmentation, or constraints on market access.

This distinction is central to the dissertation's theoretical framework. Risk and strategic importance are not automatic market determinants. They are sources of investor interpretation. The classification identifies the type of information an event is likely to contain, while the event-study results show how markets appear to respond. Deviant cases are therefore theoretically important rather than merely inconvenient. Joint Sword-2024B is important because it is classified as a risk event yet produces positive abnormal returns in the current results. BIS Advanced Semiconductor Export Controls is important because it is classified as of strategic importance yet produces negative abnormal returns. These cases help evaluate whether the adaptation and interpretation framework is more persuasive than a simple risk model.

4.5 Event Study Methodology

An event study examines how financial markets move around a specific event date. The basic logic is that if an event provides new information relevant to investors, prices may adjust around the date when that information becomes public. This approach is well suited to the dissertation because Taiwan-related geopolitical and strategic events can often be dated precisely. The Pelosi Visit, Joint Sword exercises, export controls, and TSMC investment announcements all have identifiable event dates.

The dissertation examines short event windows around each event. The main windows are Day 0, $[-1, +1]$, $[-3, +3]$, and $[-7, +7]$. Day 0 captures the immediate event-day reaction. The $[-1, +1]$ window allows for small timing differences, such as events announced after market close or responses that occur across adjacent trading days. The $[-3, +3]$ window captures somewhat broader short-run adjustment. The $[-7, +7]$ window is the principal window for this dissertation because Taiwan-related events often unfold over several trading days, especially when diplomatic actions are followed by PRC military responses or official statements.

The analysis uses both raw returns and abnormal returns. Raw returns show how the TWSE and TSMC moved during the event window. Abnormal returns adjust these movements against broader market benchmarks. In the current design, TWSE returns are adjusted against NASDAQ returns, while TSMC returns are adjusted against the Philadelphia Semiconductor Index. The logic is straightforward: if the whole technology market rises or falls during the same period, raw Taiwan returns may exaggerate or understate the Taiwan-specific component. Benchmark adjustment therefore helps separate Taiwan-related movement from broader market movement, although it cannot fully solve the identification problem.

The abnormal returns used here are calculated through simple benchmark subtraction rather than a fully estimated market model. TWSE abnormal return is the TWSE daily return minus the NASDAQ daily return. TSMC abnormal return is the TSMC daily return minus the SOX daily return. Cumulative abnormal returns are then calculated by summing daily abnormal returns over the event window. This method is accessible, transparent, and appropriate for a preliminary dissertation-level event study. It also has limitations. It assumes that the benchmark captures relevant market-wide movement and does not estimate asset-specific beta or normal-return behaviour.

The event-study results should therefore be interpreted cautiously. A positive abnormal return does not prove that investors adapted, and a negative abnormal return does not prove that an event caused market damage. Rather, the results indicate whether observed movements are consistent with particular interpretations. If risk events are negative on average, this supports the threat expectation. If repeated risk events become less negative over time, this is consistent with adaptation, provided rival explanations are considered. If strategic-importance events are positive on average, this supports the opportunity expectation. The method provides evidence for evaluating theoretical consistency, not definitive causal proof.

4.6 Variables and Operationalisation

The dependent variable is financial-market reaction. It is operationalised through daily returns, Day 0 abnormal returns, and cumulative abnormal returns for TWSE and TSMC. The TWSE represents the broader Taiwan equity market and is used to capture market-wide Taiwan exposure. TSMC represents Taiwan's strategic semiconductor position and is used to capture firm-level exposure to technology competition, supply-chain politics, and AI-related expectations.

The main independent variable at the theoretical level is grey-zone sovereignty. Because grey-zone sovereignty is a structural condition rather than a daily numerical variable, it is operationalised through the event sample it generates. The observable events are the empirical manifestations of Taiwan's unresolved status and strategic economic centrality. In this sense, the event dataset is the bridge between the theoretical condition and the market outcomes.

The main event-level variables are mechanism, event category, event repetition, and surprise level. Mechanism distinguishes Risk from Strategic_Importance. Event category identifies the substantive form of the event. Event repetition measures the degree to which similar events have occurred previously, especially within repeated military-demonstration families. Surprise level measures the extent to which an event was anticipated before Day 0. The dissertation uses surprise cautiously because low surprise may indicate that a muted market reaction reflects anticipation rather than adaptation.

Adaptation is not directly observed. It is inferred from patterns consistent with learning and reduced novelty. The clearest theoretical expectation is that repeated grey-zone shocks should become less damaging over time if investors learn to interpret them as familiar or contained. This inference is strengthened when repeated events are comparable, when later

events produce weaker negative abnormal returns, and when the pattern remains visible after benchmark adjustment. It is weakened if the same pattern can be explained by prior anticipation, AI-sector momentum, semiconductor-cycle strength, or benchmark choice.

The operationalisation therefore reflects the cautious logic of the dissertation. Investor interpretation is the immediate mechanism producing market reaction, while adaptation is the deeper mechanism that may change interpretation over time. The empirical analysis does not observe investors interpreting events in real time. It observes market prices and evaluates whether the observed patterns align with the theoretical expectations.

4.7 Rival Explanations

The first rival explanation is anticipation. Many Taiwan-related events are partly predictable. Military responses may follow diplomatic visits, inaugurations, speeches, or U.S.-Taiwan interactions. If investors expected a PRC response before Day 0, market adjustment may have occurred before the event window. A muted or positive event-window reaction would then reflect prior pricing rather than adaptation. This is the most serious rival explanation because repeated shocks are often easier to anticipate. The dissertation addresses this by using pre-event windows and by treating surprise coding as a necessary control rather than as evidence of adaptation.

The second rival explanation is the AI boom. The sample period overlaps with a major surge in investor enthusiasm around artificial intelligence, advanced chips, and high-performance computing. TSMC is closely connected to this ecosystem. Positive abnormal returns around TSMC or Taiwan technology events may therefore reflect AI-related optimism rather than geopolitical interpretation. This is particularly relevant for later events and for cases where TSMC outperforms the broader Taiwan market.

The third rival explanation is the semiconductor cycle. Semiconductor firms move through cycles of demand, inventory adjustment, capital expenditure, and earnings expectations. TSMC may outperform or underperform because of sector-specific conditions rather than because investors are interpreting Taiwan-related geopolitical events differently. The use of SOX as a benchmark partially addresses this issue, but it does not eliminate it. A stronger future design would compare TSMC with Samsung, SK Hynix, NVIDIA, and other relevant semiconductor peers.

The fourth rival explanation is benchmark specification. The current benchmark choices are defensible but not definitive. NASDAQ is a U.S. technology-heavy index rather than a Taiwan-specific counterfactual. SOX captures semiconductor-sector movement, but it may also be affected by the same AI and technology shocks that affect TSMC. Different benchmarks, such as MSCI Asia, MSCI Emerging Markets, regional technology indices, or firm-level peers, could change the size or sign of abnormal returns. This matters because the raw versus abnormal comparison already shows that adjustment method affects interpretation.

The fifth rival explanation is event-selection bias. The event sample is manually constructed and may overrepresent visible, headline-grabbing military and diplomatic events.

Strategic-importance events are fewer in number, which makes their average returns sensitive to individual cases. Event clusters may also overweight particular crisis periods. The dissertation addresses this by documenting inclusion and exclusion rules, identifying overlap clusters, and distinguishing the broader calculated sample from the overlap-reduced sample. Nevertheless, event selection remains a significant limitation.

These rival explanations do not invalidate the research design. They define the conditions under which the adaptation interpretation should be treated cautiously. If repeated risk events become less negative even after accounting for anticipation, benchmarks, AI-sector movement, semiconductor cycles, and event selection, confidence in the adaptation interpretation would increase. If the pattern disappears under these checks, the adaptation argument would need to be revised.

4.8 Limitations

The first limitation is sample size. The final overlap-aware sample is necessarily small because Taiwan-related geopolitical shocks cluster and because the study requires complete market data around each event. This limits statistical inference and makes category averages sensitive to individual observations. The dissertation therefore uses the event study as a structured descriptive method rather than as a basis for strong causal claims.

The second limitation is event-window overlap. Taiwan-related crises often unfold over several days or weeks. Diplomatic visits may be preceded by warnings and followed by military exercises, sanctions, or trade restrictions. Even with overlap reduction, it is difficult to isolate a single information shock. The event-study windows should therefore be interpreted as capturing market reactions around episodes rather than perfectly isolated events.

The third limitation is market-data and calendar alignment. Taiwan and U.S. markets do not always trade on the same days, and event dates sometimes fall on weekends or holidays. The study uses the nearest available trading dates where necessary, but this introduces timing uncertainty. The $[-7, +7]$ window helps reduce some of this problem, but it cannot eliminate it.

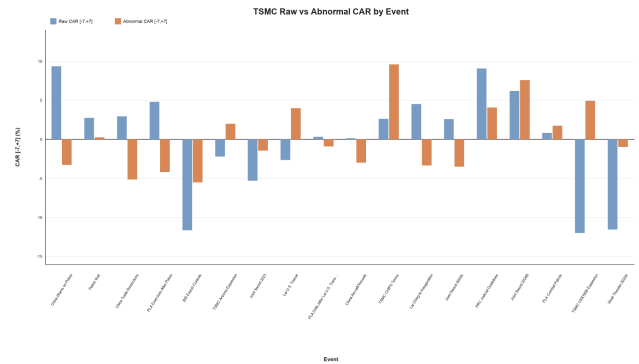
The fourth limitation is the abnormal-return model. Simple benchmark subtraction is transparent but limited. It does not estimate normal returns, beta, volatility, or asset-specific sensitivity to benchmarks. It also relies on the assumption that NASDAQ and SOX are appropriate controls. Future work should test alternative benchmarks and market-model specifications.

The fifth limitation is measurement of adaptation. Adaptation is inferred rather than directly observed. The dissertation does not have survey evidence, trading-desk data, analyst-expectation data, or direct measures of investor beliefs. It infers adaptation only when repeated shocks appear to become less damaging in a way that is consistent with the theory. This inference remains vulnerable to anticipation and broader market conditions.

The final limitation is scope. The dissertation focuses on TWSE and TSMC because they represent broad Taiwan equity exposure and strategic semiconductor exposure. This is theoretically justified, but it excludes other relevant assets such as USD/TWD, credit default

swaps, options-implied volatility, sectoral indices, and foreign-exposed Taiwan firms. A fuller study would incorporate additional asset classes to examine whether adaptation appears across markets or only in equities.

These limitations reinforce the dissertation's cautious empirical claim. The research design does not prove investor adaptation. It evaluates whether Taiwan-related market responses are consistent with an adaptation-based interpretation of grey-zone sovereignty. This is a more defensible and theoretically useful objective than claiming to identify a definitive causal mechanism from a small event-study sample.



Chapter 5: Results

5.1 Overview of Market Responses

This chapter presents the empirical results from the Taiwan event-study analysis. Its purpose is descriptive: to report observed patterns in abnormal returns around Taiwan-related geopolitical-risk and strategic-importance events.

The analysis uses benchmark-adjusted abnormal returns for two financial-market indicators: the TWSE, representing the broader Taiwan equity market, and TSMC, representing Taiwan's strategic semiconductor exposure. TWSE abnormal returns are calculated relative to NASDAQ returns, while TSMC abnormal returns are calculated relative to the Philadelphia Semiconductor Index. The main outcome is cumulative abnormal return over the $[-7, +7]$ event window, although Day 0 and shorter windows are also reported in the underlying results.

The abnormal-return file contains eighteen events. Fourteen are coded as geopolitical-risk events and four are coded as strategic-importance events. This imbalance should be noted from the outset: risk events are more frequent in the sample, while strategic-importance events are fewer and therefore more sensitive to individual cases.

The first descriptive pattern is clear: risk events are negative on average, while strategic-importance events are positive on average. For risk events, the average TWSE CAR over the $[-7, +7]$ window is -2.647, while the average TSMC CAR is -0.571. For strategic-importance events, the average TWSE CAR is 2.974, while the average TSMC CAR is 2.754. Day 0 abnormal returns show the same broad contrast. Risk events have negative average Day 0 abnormal returns for both TWSE and TSMC, while strategic-importance events have positive average Day 0 abnormal returns. The second descriptive pattern is that benchmark adjustment changes the appearance of the results. For risk events, raw average returns are slightly positive, but abnormal returns are negative. This suggests that once broader market movements are considered, Taiwan-related underperformance becomes more visible. For strategic-importance events, raw average returns are negative, but abnormal returns are positive. This indicates that some events may look weak in raw price terms while still

outperforming broader benchmarks. The contrast between raw and abnormal returns is important because it shows why benchmark adjustment is necessary for interpretation.

Figure 5.1 would visualise the average CAR $[-7, +7]$ by event type, comparing geopolitical-risk events with strategic-importance events for both TWSE and TSMC. The purpose of the figure would be to show the main descriptive contrast in the results: risk events are negative on average, while strategic-importance events are positive on average.

The third descriptive pattern is variation. The average results conceal considerable dispersion across events. Across the full sample, the average TWSE CAR $[-7, +7]$ is negative, while the average TSMC CAR is close to zero. The range is wide: TWSE CAR varies from -8.971 to 7.391, while TSMC CAR varies from -5.499 to 9.597. This spread is central to the dissertation's puzzle. The results do not show a simple pattern in which all geopolitical-risk events produce negative abnormal returns or all strategic-importance events produce positive abnormal returns. Instead, the results show group-level tendencies combined with important exceptions.

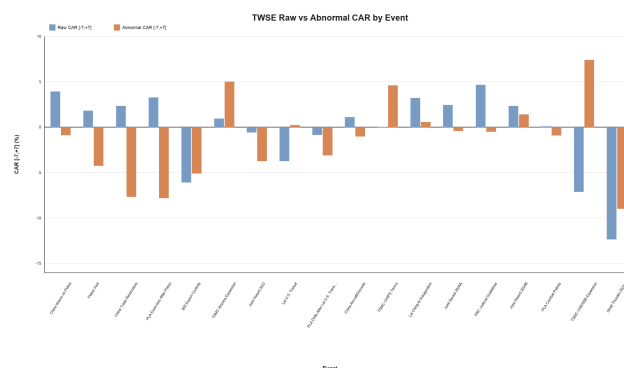
5.2 Responses to Geopolitical-Risk Events

Geopolitical-risk events show negative average abnormal returns. This is the strongest descriptive finding in the chapter. Risk events have an average TWSE CAR $[-7, +7]$ of -2.647 and an average TSMC CAR $[-7, +7]$ of -0.571. The broader Taiwan market response is more negative than the TSMC response, suggesting that country-level risk may be more visible in the TWSE than in TSMC. This pattern is consistent with the expectation that events involving coercion, escalation, diplomatic tension, or sovereignty-related pressure tend to increase uncertainty for Taiwan markets.

The sign pattern reinforces this point. Ten of fourteen risk events have negative average CAR over the $[-7, +7]$ window, while four have positive average CAR. On Day 0, eleven of fourteen risk events are negative for TWSE and eleven of fourteen are negative for TSMC. This indicates that negative market responses are common among risk events, especially on the event day. However, the presence of positive risk-event observations also shows that the pattern is not uniform.

The most negative observations are concentrated among risk events. Four of the five most negative events in the sample are coded as risk events. These include China Trade Restrictions After Pelosi Visit, PLA Military Exercises After Pelosi Visit, Strait Thunder-2025A, and Joint Sword 2023. The China Trade Restrictions event records a TWSE CAR of -7.673 and a TSMC CAR of -5.125, producing an average CAR of -6.399. The PLA Military Exercises After

(Figure 5.1 presents the average abnormal CAR $[-7, +7]$ by event type for both TWSE and TSMC.)



Pelosi Visit event records a TWSE CAR of -7.800 and a TSMC CAR of -4.189, with an average CAR of -5.995. Strait Thunder-2025A records the most negative TWSE CAR in the sample, at -8.971, although its TSMC CAR is less negative at -0.947.

mechanism	n	negative average CAR [-7,+7]	positive average CAR [-7,+7]	negative TWSE Day 0	positive TWSE Day 0	negative TSMC Day 0	positive TSMC Day 0
Risk	14	10	4	11	3	11	3
Strategic_Importance	4	1	3	1	3	1	3

Figure 5.2 would display event-level average CAR [-7,+7] for all events, ordered from most negative to most positive. This figure would make visible the wide dispersion hidden by group averages and would highlight the main deviant observations, including Joint Sword-2024B and BIS Advanced Semiconductor Export Controls.

event date	event name	mechanism	TWSE CAR [-7,+7]	TSMC CAR [-7,+7]	average CAR [-7,+7]
2022-08-03	China Trade Restrictions After Pelosi Visit	Risk	-7.673	-5.125	-6.399
2022-08-04	PLA Military Exercises After Pelosi Visit	Risk	-7.800	-4.189	-5.995
2022-10-07	BIS Advanced Semiconductor Export Controls	Strategic_Importance	-5.088	-5.499	-5.294
2025-04-01	Strait Thunder-2025A	Risk	-8.971	-0.947	-4.959
2023-04-08	Joint Sword 2023	Risk	-3.728	-1.416	-2.572

(Figure 5.2: Event-level Average CAR [-7,+7])

The Pelosi-related crisis period is especially important descriptively. The China Trade Restrictions event and the PLA Military Exercises event both appear among the most negative observations. Pelosi Visit itself is also negative for TWSE over the [-7,+7] window, with a TWSE CAR of -4.233, although TSMC CAR is slightly positive at 0.234. This suggests that the broader Taiwan market registered a negative abnormal response during the Pelosi crisis period, while TSMC's response was less uniformly negative.

Military demonstrations also vary. Joint Sword 2023 is negative for both TWSE and TSMC over the [-7,+7] window. Joint Sword-2024A is negative for both indicators as well, although the TWSE decline is small at -0.426 while TSMC falls by -3.469. By contrast, Joint Sword-2024B is positive over the same window for both TWSE and TSMC. This variation within the military-demonstration category is one of the most important descriptive findings. It shows that the event category alone does not fully determine market direction.

The results for political-signalling and diplomatic-risk events are also mixed. Lai U.S. Transit is coded as a risk event but produces positive CAR over the [-7,+7] window for both TWSE and TSMC. PRC Judicial Guidelines on Taiwan Independence produces a slightly negative TWSE CAR but a positive TSMC CAR. These observations do not overturn the average negative pattern among risk events, but they show that the market response to geopolitical risk is heterogeneous.

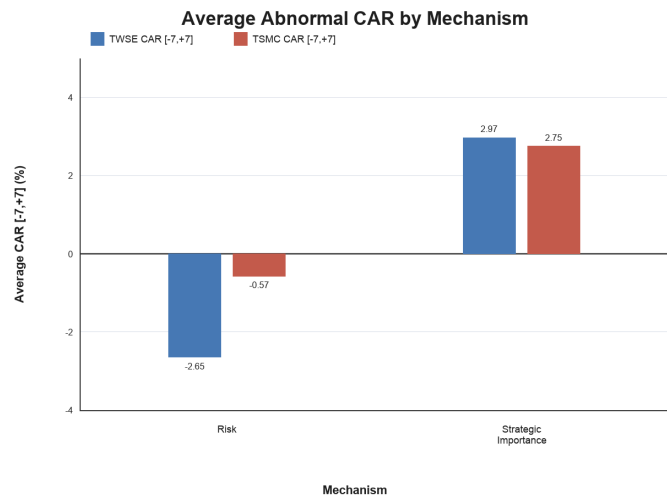
5.3 Responses to Strategic-Importance Events

Strategic-importance events are positive on average. The four events in this category have an average TWSE CAR $[-7, +7]$ of 2.974 and an average TSMC CAR of 2.754. Day 0 abnormal returns are also positive on average. This pattern is consistent with the expectation that events highlighting Taiwan's role in semiconductor geopolitics, supply-chain restructuring, and technology competition may generate positive market responses when investors interpret them as signals of policy support, strategic value, or future investment.

Three of the five most positive events in the sample are strategic-importance events. TSMC CHIPS Act Preliminary Terms is the most positive observation overall, with a TWSE CAR of 4.595 and a TSMC CAR of 9.597. TSMC US\$165B U.S. Expansion is the second most positive event, with a TWSE CAR of 7.391 and a TSMC CAR of 4.935. TSMC Arizona US\$40B Expansion is also positive, with a TWSE CAR of 4.998 and a TSMC CAR of 1.985. These results suggest that TSMC-related strategic investment and U.S. industrial-policy support are associated with positive abnormal returns in the current sample.

The strongest strategic-importance event is the TSMC CHIPS Act Preliminary Terms announcement. It produces positive abnormal returns for both TWSE and TSMC, with the TSMC response particularly large. This suggests that markets responded favourably to an event associated with industrial-policy support and supply-chain restructuring. The TSMC US\$165B U.S. Expansion event also produces positive abnormal returns for both indicators. The positive TWSE response indicates that strategic-importance events may affect the broader Taiwan market as well as TSMC.

However, the category is not uniformly positive. BIS Advanced Semiconductor Export Controls is a strategic-importance event but appears among the five most negative observations. It records a TWSE CAR of -5.088 and a TSMC



(*Figure 5.3: Repetition Sequence from Pelosi Visit to Joint Sword-2024B)

Figure 5.3. Illustrative Adaptation Sequence

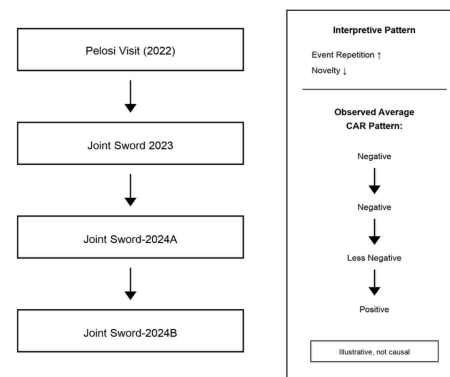


Figure 5.3. Illustrative Adaptation Sequence.

Note: The sequence shows how repeated Taiwan-related military or crisis events appear less uniformly negative over time. The figure is intended as a descriptive illustration rather than a statistical test of adaptation.

CAR of -5.499. This is a crucial descriptive result because it prevents an overly simple interpretation of strategic importance. Events connected to semiconductor geopolitics can be positive when they signal support, investment, or protection. They can be negative when they signal restriction, fragmentation, compliance costs, or uncertainty in technology supply chains.

The strategic-importance results therefore support a conditional descriptive pattern. The category is positive on average, but not because strategic importance automatically produces positive returns. Rather, the positive average is driven by TSMC-related investment and policy-support events, while the BIS export-control event shows that technology competition can also be market-negative. This distinction matters for the dissertation's theoretical framework because it supports the claim that strategic importance is a source of interpretation rather than a deterministic market signal.

5.4 Deviant Cases

The results include several deviant cases, meaning observations where market reactions contradict the broad expectation associated with the event classification. These cases are important because they reveal the limits of a simple risk-versus-opportunity classification. They do not invalidate the classification scheme, but they show that market responses vary within each category.

The key deviant case is Joint Sword-2024B. It is classified as a risk event because it was a military demonstration around Taiwan. Under a simple geopolitical-risk expectation, such an event would be expected to produce negative abnormal returns. Instead, Joint Sword-2024B records a positive TWSE CAR of 1.390 and a strongly positive TSMC CAR of 7.561 over the [-7,+7] window. Its average CAR of 4.475 places it among the five most positive observations in the sample. This makes it the most important case for the dissertation's central puzzle: a high-profile military event did not correspond to negative abnormal returns in the observed window.

The deviance of Joint Sword-2024B is especially clear when compared with earlier military-demonstration events. Joint Sword 2023 records negative CAR for both TWSE and TSMC. Joint Sword-2024A records a slightly negative TWSE CAR and a more negative TSMC CAR. The PLA Military Exercises After Pelosi Visit event is strongly negative for both indicators. Joint Sword-2024B therefore differs from several earlier military-risk events. The chapter does not explain this difference causally; it simply notes that the observation is inconsistent with a uniform risk-response model.

The second major deviant case is BIS Advanced Semiconductor Export Controls. It is classified as a strategic-importance event because it belongs to semiconductor geopolitics and U.S.-China technology competition. A broad strategic-importance expectation might suggest positive returns if investors interpreted the event as highlighting the value of advanced semiconductor capabilities. Instead, the event produces negative abnormal returns for both TWSE and TSMC. This indicates that strategic-importance events can also be associated with

negative market responses, especially when the event involves restrictions rather than investment or support.

Lai U.S. Transit is another deviant case, though less central than Joint Sword-2024B. It is classified as a risk event but produces positive CAR for both TWSE and TSMC. PRC Judicial Guidelines on Taiwan Independence also shows divergence across indicators: TWSE is slightly negative while TSMC is positive. These cases show that not all risk events are priced in the same way and that TSMC may respond differently from the broader Taiwan market.

The deviant cases are useful because they sharpen rather than weaken the empirical contribution. If all risk events were negative and all strategic-importance events were positive, the results would support a simple classification model. Instead, the data show average tendencies with important exceptions. The exceptions are precisely what motivate the theoretical framework developed in Chapters 2 and 3. At this stage, however, the chapter confines itself to description. The implications of these deviant cases for adaptation, anticipation, and rival explanations are developed in Chapter 6.

5.5 Preliminary Evidence Consistent with Adaptation

The results contain limited preliminary evidence compatible with an adaptation-based interpretation. This evidence should be treated cautiously. Adaptation is not directly observed in the data. The analysis observes abnormal returns around repeated events and asks whether the pattern is consistent with the possibility that investors interpret later shocks as more familiar or less novel. It does not prove that investors learned, adapted, or consciously reclassified Taiwan-related risk.

The clearest descriptive comparison involves four core events for which repetition and surprise coding are currently available: Pelosi Visit, Joint Sword 2023, Joint Sword-2024A, and Joint Sword-2024B. These events form a sequence of Taiwan-related crisis or military-demonstration episodes. In this sequence, event repetition rises from 1 for Pelosi Visit to 4 for Joint Sword-2024B. The average CAR over the $[-7, +7]$ window is negative for the first three events and positive for Joint Sword-2024B.

Figure 5.3 would plot this repetition sequence, showing event repetition on the horizontal axis and average CAR $[-7, +7]$ on the vertical axis. Its purpose would be illustrative rather than confirmatory: the figure would show the movement from negative average CAR in the earlier

Figure 5.4. Illustrative Adaptation Sequence

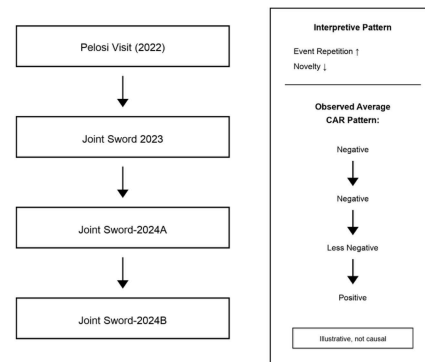


Figure 5.4. Illustrative Adaptation Sequence.

Note: The sequence illustrates how repeated Taiwan-related military or crisis events appear less uniformly negative over time. The figure is descriptive and should not be interpreted as a statistical test of adaptation.

observations to positive average CAR for Joint Sword-2024B, while also making clear that the sequence contains only four coded cases.

This sequence is consistent with, but does not prove, an adaptation interpretation. Pelosi Visit and Joint Sword 2023 are negative on average. Joint Sword-2024A remains negative on average but less strongly so. Joint Sword-2024B becomes positive on average. The sequence should be interpreted as an illustrative pattern rather than a statistically robust trend. It suggests that later repeated military demonstrations may have produced less negative market responses than earlier crisis events, but the sequence is not perfectly monotonic. Joint Sword 2023 is more negative than Pelosi Visit on average, and Joint Sword-2024A still shows a negative TSMC response. The number of coded observations is also very small.

The surprise coding further complicates interpretation. Pelosi Visit has a higher surprise level than the later Joint Sword events. This is consistent with the idea that novelty matters: the Pelosi episode may have carried more information because it was more surprising, while later Joint Sword events may have been more anticipated. Yet this creates a rival interpretation. A weaker reaction to later events may reflect anticipation rather than adaptation. If investors expected a PRC military response before Day 0, the event-window return may understate the full market adjustment.

For this reason, the evidence should be described as consistent with adaptation rather than as evidence of adaptation itself. The data show that later repeated events do not always produce the same negative response as earlier events. They also show that Joint Sword-2024B is a positive deviant case within the risk category. These patterns are compatible with the adaptation-based framework, but they also fit other possible explanations, including anticipation, AI-sector momentum, semiconductor-cycle effects, benchmark specification, and event-selection choices. Chapter 6 evaluates these rival explanations more directly.

5.6 Summary of Findings

The chapter reports six main empirical findings. First, geopolitical-risk events are negative on average. This is visible in both Day 0 abnormal returns and cumulative abnormal returns over the $[-7, +7]$ window. The broader Taiwan market, represented by TWSE, shows a stronger negative average response than TSMC.

Second, strategic-importance events are positive on average. TSMC CHIPS Act Preliminary Terms, TSMC US\$165B U.S. Expansion, and TSMC Arizona US\$40B Expansion all produce positive abnormal returns over the $[-7, +7]$ window. These events are connected to semiconductor policy, supply-chain restructuring, and strategic investment.

Third, the results show substantial variation across events. Most risk events are negative, but four of fourteen risk events have positive average CAR over the $[-7, +7]$ window. Most strategic-importance events are positive, but BIS Advanced Semiconductor Export Controls is negative. The results therefore do not support a uniform one-direction model.

Fourth, Joint Sword-2024B is the key deviant case. It is classified as a geopolitical-risk event because it is a military demonstration, yet it produces positive abnormal returns over the $[-7, +7]$ window, especially for TSMC. This result is central to the dissertation's empirical puzzle.

Fifth, TWSE and TSMC do not always move together. Some events are negative for the broader Taiwan market but less negative or positive for TSMC. This divergence suggests that broad-market Taiwan risk and semiconductor-specific expectations may differ.

Sixth, the repetition sequence from Pelosi Visit to Joint Sword-2024B provides preliminary evidence compatible with an adaptation-based interpretation. The evidence is limited and should not be treated as proof. It shows that later repeated military events are not uniformly more damaging and that the most recent Joint Sword event in the sequence is positive over the main event window.

The results are consistent with the theoretical framework developed in Chapters 2 and 3, but they do not prove it. The empirical pattern is best described as conditional. Risk events tend to be negative, strategic-importance events tend to be positive, and substantial variation remains within both categories. The descriptive evidence suggests that a simple geopolitical-risk model may be insufficient to explain all observed market reactions. Chapter 6 therefore turns from description to interpretation, assessing whether these patterns are better explained by adaptation or by alternative explanations such as anticipation effects, AI-sector optimism, semiconductor-cycle dynamics, benchmark specification, and event-selection choices.

Chapter 6: Discussion

6.1 Revisiting the Puzzle

This dissertation began from a puzzle: why do some Taiwan-related geopolitical shocks fail to generate persistent market damage despite elevated objective risk? Taiwan is a highly visible geopolitical flashpoint. Its contested sovereignty, exposure to Chinese military pressure, role in U.S.-China competition, and position in global semiconductor supply chains all suggest that Taiwan-related political shocks should matter for financial markets. The results in Chapter 5 support this general expectation, but they also complicate it.

The main empirical pattern is conditional rather than uniform. Geopolitical-risk events are negative on average. Strategic-importance events are positive on average. However, substantial variation exists within both categories. Some risk events are associated with positive abnormal returns, while one strategic-importance event, BIS Advanced Semiconductor Export Controls, is associated with negative abnormal returns. The key deviant case is Joint Sword-2024B: a military-demonstration event that would normally be expected to heighten geopolitical risk, yet is associated with positive abnormal returns in the main event window, especially for TSMC.

The purpose of this chapter is to evaluate possible explanations for this pattern. Chapter 5 described what happened. This chapter asks how the pattern should be interpreted. The analysis does not treat adaptation as an established fact. Instead, it

evaluates whether the results are consistent with an adaptation-based interpretation and compares that interpretation with plausible rivals. This distinction is important because the evidence is suggestive but not definitive. Market prices are observable; investor beliefs are not. Any discussion of adaptation must therefore remain inferential and cautious.

The central interpretive question is whether Taiwan's grey-zone sovereignty produces recurring shocks that investors gradually learn to price differently. The results are consistent with this possibility, but they do not exclude other explanations. Anticipation effects, semiconductor-cycle dynamics, AI-sector optimism, benchmark specification, and event-selection choices could all affect the observed abnormal returns. The contribution of this chapter is therefore not to declare a winner among explanations, but to clarify what each explanation can and cannot account for.

6.2 Evaluating the Adaptation Interpretation

The adaptation interpretation follows directly from the theoretical framework developed in Chapter 3. Grey-zone sovereignty is a persistent condition rather than a one-off crisis. Because Taiwan's status remains unresolved, Taiwan-related shocks recur. Military exercises, diplomatic protests, sovereignty-related speeches, U.S.-Taiwan interactions, and technology-policy events are repeated manifestations of the same underlying political ambiguity. If investors observe similar shocks over time, they may begin to distinguish between events that materially alter the risk environment and events that belong to a familiar repertoire of grey-zone signalling.

The most relevant empirical sequence is Pelosi Visit, Joint Sword 2023, Joint Sword-2024A, and Joint Sword-2024B. These events are not identical, but they form a recognisable sequence of Taiwan-related crisis or military-demonstration episodes. Pelosi Visit represents an earlier and more surprising diplomatic shock followed by major PRC pressure. Joint Sword 2023 followed the Tsai-McCarthy meeting and involved large-scale military exercises. Joint Sword-2024A followed Lai Ching-te's inauguration. Joint Sword-2024B followed another sovereignty-related political trigger and again involved military activity around Taiwan.

The sequence is compatible with adaptation because later events appear less uniformly negative than earlier ones. Pelosi Visit and Joint Sword 2023 are negative on average over the main event window. Joint Sword-2024A remains negative on average, but less strongly. Joint Sword-2024B is positive on average. The sequence should be interpreted as an illustrative pattern rather than a robust test of adaptation. It is not fully monotonic, and the number of observations is too small to support strong inference. Nevertheless, it is consistent with the possibility that repeated exposure reduced the novelty of later military demonstrations. If investors came to view some PLA exercises as recurring grey-zone signals rather than new escalatory breaks, their market response may have weakened.

Adaptation in this sense involves learning, expectation formation, and reduced novelty. Learning does not mean that investors believe geopolitical risk has disappeared. It means that investors may become better able to classify different types of Taiwan-related events. A novel diplomatic crisis may be priced as a larger threat because its implications are uncertain. A repeated military demonstration may be priced differently if it resembles earlier episodes and does not appear to change expected fundamentals. The same objective event type can therefore carry different informational content depending on its place in a sequence.

Joint Sword-2024B is important because it is the strongest challenge to a simple risk model. It is classified as a geopolitical-risk event because it was a military demonstration around Taiwan. Yet it appears among the most positive observations in the current results. This does not show that investors adapted. It does, however, provide a case where the adaptation interpretation has possible explanatory leverage. If a repeated military-demonstration event is not priced as damaging, one possible explanation is that investors interpreted it as a familiar, anticipated, or contained grey-zone signal.

The adaptation interpretation remains limited. First, adaptation is inferred rather than directly observed. The dissertation does not have survey evidence, analyst-expectation data, trading-desk evidence, or direct measures of investor beliefs. Second, the repetition sequence includes only four coded observations. Third, later events occurred in different market environments, especially during a period of strong technology-sector dynamics. Fourth, repeated events are also more likely to be anticipated, which makes adaptation difficult to separate from prior pricing. The correct conclusion is therefore cautious: the evidence is compatible with adaptation but cannot establish it as the only or necessarily dominant explanation.

6.3 Threat, Opportunity, and Investor Interpretation

Adaptation alone is insufficient to explain the results. The dissertation's framework does not argue that all market variation comes from repeated exposure. Rather, it argues that investor interpretation is the immediate mechanism linking events to market reactions. Adaptation changes interpretation over time, but it is not the only source of interpretation. Events may also be interpreted as threats or opportunities depending on their content.

Threat interpretation is the most direct channel for geopolitical-risk events. Military escalation, coercive economic measures, diplomatic shocks, and sovereignty-related political pressure can all raise perceived uncertainty. They may increase concerns about conflict risk, supply-chain disruption, investor confidence, or future policy instability. The negative average abnormal returns for risk events are consistent with this threat channel. The Pelosi crisis period, Joint Sword 2023, and Strait Thunder-2025A all fit the expectation that acute risk events can produce negative market responses.

The threat channel matters because it prevents the dissertation from overstating adaptation. If risk events were not negative on average, the project might imply that markets have largely discounted Taiwan-related geopolitical risk. The results do not support that claim. Objective risk remains important. Many events associated with military pressure or coercion are associated with negative abnormal returns. The puzzle is not that risk has no effect. The puzzle is that risk does not have a uniform effect.

Opportunity interpretation operates differently. Strategic-importance events may be interpreted as evidence of Taiwan's centrality to semiconductor supply chains, U.S. industrial policy, economic security, or long-run technology demand. CHIPS Act-related events and TSMC expansion announcements are associated with positive abnormal returns in the current sample. These events may signal state support, investment, supply-chain protection, or the strategic value of advanced semiconductor capacity. In these cases, Taiwan's geopolitical position may be read not only as vulnerability but also as economic significance.

However, opportunity interpretation is also conditional. BIS Advanced Semiconductor Export Controls is a strategic-importance event but is associated with negative abnormal returns. This suggests that strategic importance can generate either positive or negative interpretation depending on the event. Technology competition may imply policy support and investment, but it may also imply restriction, fragmentation, compliance costs, or uncertainty. Strategic importance is therefore not a synonym for market optimism.

The distinction between adaptation and interpretation is central. Interpretation is the immediate mechanism: investors price an event as threat, opportunity, constraint, familiar signal, or surprise. Adaptation is a deeper mechanism that may change interpretation over time. A repeated military event may still be threatening, but less threatening than a novel shock if investors view it as part of an established grey-zone pattern. A strategic-importance event may be positive if it signals support, but negative if it signals restriction. The results therefore point to a layered framework in which geopolitical risk and strategic importance provide the content of events, adaptation shapes how repeated events are read, and interpretation produces the market reaction.

This connects directly to Chapter 3. The theoretical framework proposed that Taiwan's grey-zone sovereignty generates recurring shocks, that repeated shocks may produce adaptation, and that investor interpretation mediates market response. The empirical results are compatible with this framework because they show average differences between risk and strategic-importance events, alongside deviant cases that require interpretation. Yet the framework remains provisional. It is a way to organise the patterns, not a definitive causal account.

6.4 Rival Explanations

6.4.1 Anticipation Effects

Anticipation is the most serious rival explanation for the adaptation interpretation. Many Taiwan-related events are not complete surprises. Diplomatic visits are often discussed in advance. Inaugurations and major speeches are scheduled. PRC responses to Taiwan-related political events may be expected. Military exercises following high-profile diplomatic or sovereignty-related triggers may therefore contain less new information than their objective severity suggests.

This explanation is plausible because event studies measure market movement around a defined window. If investors adjust before Day 0, then the observed event-window reaction may appear muted or positive even though the event affected expectations earlier. This is especially relevant for later Joint Sword events. If investors expected a PRC military response after Lai's inauguration or National Day speech, the response may have been partly priced before the formal military event date. In that case, weak or positive abnormal returns would not necessarily indicate adaptation. They could reflect prior pricing.

Anticipation can explain why repeated events may appear less damaging. Repetition makes events more predictable. If the market expects a familiar pattern of PRC signalling, the event date itself may contain little new information. Anticipation can also explain why surprise coding matters: Pelosi Visit appears more surprising than later Joint Sword events, and the later events may therefore produce weaker immediate reactions.

However, anticipation cannot fully explain the whole pattern by itself. It explains muted responses to expected risk events, but it does not explain why strategic-importance events are positive on average or why BIS export controls are negative. It also overlaps with adaptation rather than replacing it entirely. Adaptation concerns how repeated exposure changes the meaning of a class of events. Anticipation concerns whether a particular event was expected before Day 0. A repeated event can be both anticipated and interpreted as familiar. Future work must separate these mechanisms more clearly, but the present results cannot do so decisively.

6.4.2 Semiconductor-Cycle Dynamics

Semiconductor-cycle dynamics provides another plausible rival explanation, especially for TSMC. Semiconductor firms are affected by demand cycles, inventory adjustments, capital expenditure, utilisation rates, customer orders, and expectations about future technology demand. TSMC may move differently from the broader Taiwan market because of firm-specific or sector-specific conditions that are not primarily geopolitical.

This explanation is plausible because several of the strongest positive observations involve TSMC or semiconductor-related events. If TSMC was benefiting from favourable

sector conditions, then positive abnormal returns around strategic-importance events may partly reflect industry fundamentals rather than investor interpretation of Taiwan's geopolitical position. The same issue applies to Joint Sword-2024B, where the strongly positive TSMC CAR contributes substantially to the event's positive average. A semiconductor-cycle explanation could account for why TSMC performs better than expected even during a geopolitical-risk event.

Semiconductor-cycle dynamics can explain asset-level divergence between TWSE and TSMC. The broader Taiwan market may respond to country-level risk, while TSMC may respond to expectations about advanced-node demand, customer relationships, or sector leadership. This helps explain why some events are negative for TWSE but less negative or positive for TSMC.

However, semiconductor-cycle dynamics do not fully explain the entire pattern. They are most relevant to TSMC, but the results also show patterns in TWSE. Some strategic-importance events are positive for both TWSE and TSMC, suggesting that the response may not be purely firm-specific. Semiconductor-cycle dynamics also do not explain why some risk events are strongly negative for the broader market. The rival explanation is therefore important but partial. It may account for some TSMC-specific positive observations, but it does not replace the need to interpret Taiwan-related political risk.

6.4.3 AI Boom and Technology Optimism

The AI boom and broader technology-sector optimism are closely related to semiconductor-cycle dynamics but deserve separate attention. The sample period overlaps with strong enthusiasm around artificial intelligence, advanced chips, data-centre investment, and high-performance computing. TSMC is central to this ecosystem. Positive returns around TSMC-linked events may therefore reflect global AI optimism rather than Taiwan-specific geopolitical interpretation.

This explanation is plausible because several positive observations are connected to TSMC, CHIPS Act support, and U.S. expansion. These events may have been interpreted not only as signs of Taiwan's strategic importance, but also as signals of growth in AI-related demand. If investors were especially optimistic about AI infrastructure and advanced semiconductors, TSMC could experience positive abnormal returns even during periods of geopolitical tension.

AI-sector optimism can also help explain why Joint Sword-2024B is positive, especially for TSMC. If TSMC was benefiting from broader technology optimism during the event window, then the positive TSMC response may not reflect adaptation to Taiwan-related military risk. It may reflect sector momentum that overwhelmed or offset the geopolitical signal.

However, AI optimism cannot fully explain all findings. It does not explain why risk events are negative on average, especially for TWSE. It also does not explain the negative

response to BIS Advanced Semiconductor Export Controls, which is part of semiconductor geopolitics but appears among the most negative observations. If AI optimism alone were driving technology-related responses, one would expect technology-linked events to be more uniformly positive. The evidence instead suggests that technology optimism may be one influence among several, not a complete explanation.

6.4.4 Benchmark Specification

Benchmark specification is a methodological rival explanation. The dissertation calculates TWSE abnormal returns relative to NASDAQ and TSMC abnormal returns relative to SOX. These benchmarks are defensible as first-pass controls because NASDAQ captures broader technology-market movement and SOX captures semiconductor-sector movement. However, they are not the only possible benchmarks, and different choices could change the results.

This explanation is plausible because the raw and abnormal returns differ meaningfully. For strategic-importance events, raw average CAR is negative while abnormal CAR is positive. This means that the interpretation of results depends on benchmark adjustment. If an alternative benchmark were used, some abnormal returns might be smaller, larger, or different in sign. NASDAQ may be too U.S.-centric and technology-heavy to represent general global market conditions. SOX may be too closely tied to the same semiconductor forces that influence TSMC. Regional Asian benchmarks, global equity indices, MSCI Taiwan, or firm-level peer comparisons could produce different conclusions.

Benchmark specification can explain why some events look different after adjustment. It can also explain why TSMC appears to outperform or underperform depending on the comparator. This is especially relevant for strategic-importance events, where the shift from negative raw returns to positive abnormal returns is substantively important.

However, benchmark specification cannot fully explain the substantive pattern by itself. The fact that benchmark choice matters does not mean the observed abnormal returns are meaningless. It means they should be interpreted as conditional on the chosen model. Benchmark concerns also do not eliminate the substantive contrast between risk and strategic importance; rather, they require testing whether that contrast survives alternative specifications. Future robustness checks using alternative benchmarks would strengthen confidence in the findings, especially if the broad risk-versus-strategic-importance pattern remains visible across specifications. The current results should therefore be treated as informative but not definitive.

6.4.5 Event Selection Bias

Event-selection bias is another important concern. The event sample is manually constructed. This is appropriate for an exploratory dissertation, but it creates risks. Highly

visible events may be overrepresented. Borderline events may be included or excluded based on researcher judgement. Crisis clusters may receive more attention than lower-salience events. Strategic-importance events are fewer than risk events, making category averages sensitive to a small number of observations.

This explanation is plausible because event selection shapes the empirical universe. The decision to include some events and exclude others affects the average returns by category. The handling of overlap also matters. The Pelosi crisis period includes warnings, the visit itself, trade restrictions, and military exercises. Treating these as separate events increases the number of observations but may also double-count a single crisis episode. Reducing overlap improves independence but reduces sample size.

Event-selection bias can explain why some patterns may appear stronger or weaker than they actually are. If the sample overrepresents severe crisis events, risk events may look more negative. If it includes only the most visible strategic-importance events, that category may appear more positive. If deviant cases are especially salient, they may receive disproportionate interpretive weight.

However, event-selection bias does not fully explain the findings. The event-selection criteria are substantively grounded in Taiwan Strait military activity, diplomatic shocks, political signalling, economic coercion, semiconductor geopolitics, and strategic technology competition. The deviant cases are not arbitrary; Joint Sword-2024B and BIS export controls are substantively relevant events. Event-selection concerns therefore require caution, transparency, and robustness checks, but they do not make the analysis invalid. They limit the strength of inference.

6.5 Theoretical Implications

The discussion has several implications for geopolitical-risk theory. The dissertation should not be interpreted as a rejection of geopolitical-risk theory. On the contrary, risk events are negative on average, and several of the most negative observations are associated with military escalation, coercion, or crisis dynamics. The results therefore support the basic claim that geopolitical risk matters. Rather, the dissertation introduces an additional interpretive layer between geopolitical shocks and market outcomes. The theoretical refinement is that geopolitical risk does not move directly or uniformly into market reaction. The relationship suggested by the dissertation is: Geopolitical Risk → Interpretation → Market Reaction, rather than: Geopolitical Risk → Market Reaction.

This distinction matters because it places meaning between event and price. Investors do not simply observe objective risk and mechanically reprice assets. They may interpret whether an event is novel or familiar, threatening or contained, restrictive or supportive, surprising or anticipated. The same broad event category can therefore generate different market responses.

For sovereignty literature, the implication is that sovereignty ambiguity has economic consequences. Taiwan's contested status does not only shape diplomatic recognition or military risk. It also produces recurring market-relevant events. Grey-zone sovereignty generates repeated shocks because the underlying political status is unresolved. This connects sovereignty debates to observable financial-market behaviour without turning the dissertation into a legal-status argument.

For geoeconomics, the dissertation highlights the dual character of strategic importance. Taiwan's semiconductor position makes it economically valuable and politically exposed. Strategic industries may attract investment, policy support, and supply-chain protection, but they may also face restrictions, fragmentation, and regulatory uncertainty. The results are consistent with this duality: TSMC-related investment and CHIPS Act events are positive, while BIS export controls are negative.

For investor behaviour, the dissertation suggests that expectation formation and event meaning are central. Adaptation is one possible mechanism through which repeated shocks may change interpretation over time. If a class of events becomes familiar, investors may assign it less negative informational content unless it changes expected fundamentals. This is not the same as saying that investors ignore risk. It means that repeated exposure may alter how risk is processed. The overall theoretical contribution can be expressed as: Grey-Zone Sovereignty → Recurring Shocks → Adaptation → Interpretation → Market Reaction.

This framework should be read as conditional and provisional. Grey-zone sovereignty explains why Taiwan-related events recur. Recurrence creates the possibility of adaptation. Adaptation may shape interpretation over time. Interpretation is the immediate mechanism through which events become market reactions. The framework does not eliminate threat or opportunity; it organises them within a broader account of how political ambiguity becomes priced.

The framework may also have relevance beyond Taiwan, although this dissertation does not test that broader claim. Taiwan is analytically useful because sovereignty ambiguity, recurring geopolitical pressure, and strategic economic importance are unusually visible in the same case. Similar dynamics may be relevant in other politically ambiguous or contested settings where repeated geopolitical shocks become familiar to investors over time. The point is not to generalise mechanically from Taiwan, but to suggest that recurring shocks and investor adaptation may not be unique to this case. Future comparative research would be needed to assess how far the framework travels.

6.6 Discussion Summary

This chapter has evaluated the main explanations for the empirical patterns reported in Chapter 5. The adaptation interpretation is plausible but not established. The sequence from Pelosi Visit to Joint Sword-2024B is compatible with the idea that repeated grey-zone shocks may become less novel and therefore less uniformly damaging, but it

should be read as an illustrative pattern rather than a robust test. Joint Sword-2024B is especially important because it is a risk event that does not behave as a simple risk model would expect.

At the same time, rival explanations remain plausible. Anticipation effects may shift repricing before Day 0. Semiconductor-cycle dynamics may influence TSMC independently of Taiwan-related risk. AI-sector optimism may help explain positive technology-linked observations. Benchmark specification may affect abnormal-return estimates. Event-selection choices may shape the sample and the apparent strength of patterns. None of these explanations should be dismissed.

The strongest conclusion is therefore cautious. Objective geopolitical risk remains important, but investor interpretation appears central to how that risk is priced. Strategic importance also matters, but it is not uniformly positive. Adaptation may help explain why some repeated military demonstrations appear less damaging, but adaptation is inferred rather than directly observed.

Chapter 7 concludes the dissertation by returning to the research question and summarising the contribution. The conclusion does not claim that geopolitical risk no longer matters. Instead, it synthesises the dissertation's more conditional claim: under grey-zone sovereignty, the relationship between geopolitical risk and market outcomes depends on how repeated shocks are interpreted by investors.

Chapter 7: Conclusion

7.1 Revisiting the Research Question

This dissertation began with a puzzle: why do some Taiwan-related geopolitical shocks fail to generate persistent market damage despite elevated objective risk? The puzzle matters because Taiwan is widely understood as a major geopolitical flashpoint. Its contested status, exposure to Chinese military pressure, role in U.S.-China strategic competition, and central position in global semiconductor production all suggest that Taiwan-related shocks should be financially significant. A simple geopolitical-risk expectation would imply that events increasing cross-strait uncertainty should generally produce negative market responses.

The empirical findings complicate that expectation. Some events do behave as a conventional geopolitical-risk model would anticipate. The Pelosi crisis period, Joint Sword 2023, and other military or coercive events are associated with negative abnormal returns. Yet the results are not uniform. Joint Sword-2024B, despite being a military demonstration, is associated with positive abnormal returns in the main event window. Strategic-importance events are positive on average, but they too are not uniformly positive. The dissertation therefore does not ask whether geopolitical risk matters. It asks why the relationship between Taiwan-related geopolitical risk and market outcomes appears conditional rather than automatic.

The answer developed across the dissertation is that Taiwan's grey-zone sovereignty produces recurring shocks that investors must interpret. Grey-zone sovereignty refers to Taiwan's persistent condition of political ambiguity: Taiwan possesses substantial practical autonomy and

strategic economic importance, but remains diplomatically constrained and politically contested. This condition generates recurring events that contain more than one possible market meaning. They may signal threat, but they may also signal strategic importance, policy support, repetition, anticipation, or constrained escalation. The dissertation has therefore examined whether observed market responses are consistent with an adaptation-based interpretation, while remaining cautious about causal claims.

7.2 Main Findings

The first main finding is that geopolitical-risk events are negative on average. In the event-study results, risk events show negative average abnormal returns for both TWSE and TSMC over the main [-7, +7] window. This finding is important because it confirms that geopolitical risk remains relevant. Taiwan-related events involving military demonstrations, diplomatic shocks, political signalling, or coercive economic pressure cannot be treated as market-neutral. The broader Taiwan market, represented by TWSE, appears especially sensitive to these events.

The second finding is that strategic-importance events are positive on average. Events connected to TSMC investment, CHIPS Act support, and semiconductor supply-chain restructuring are associated with positive average abnormal returns in the current sample. This pattern suggests that some Taiwan-related events may be interpreted through the lens of strategic value rather than only geopolitical vulnerability. Taiwan's position in semiconductor geopolitics can therefore generate positive market expectations when events are associated with policy support, investment, or supply-chain protection.

The third finding is that substantial variation exists within both categories. Risk events are not uniformly negative, and strategic-importance events are not uniformly positive. This variation is central to the dissertation's argument. It shows that classification alone does not determine market response. Risk and strategic importance are best understood as sources of investor interpretation rather than automatic predictors of market direction.

The fourth finding is that Joint Sword-2024B is the key deviant case. It is coded as a geopolitical-risk event because it was a military demonstration around Taiwan, yet it is associated with positive abnormal returns in the event window, especially for TSMC. This case is important because it directly challenges a simple risk-response model. It does not prove adaptation, but it provides a reason to ask whether repeated grey-zone military demonstrations may be interpreted differently over time.

The fifth finding is that later repeated shocks appear less uniformly negative than earlier shocks. The sequence from the Pelosi Visit to Joint Sword 2023, Joint Sword-2024A, and Joint Sword-2024B is compatible with an adaptation-based interpretation. The sequence should not be treated as a statistically robust trend, and it does not demonstrate investor learning. However, it is consistent with the possibility that repeated exposure reduces the novelty of some grey-zone shocks and alters the way investors interpret them.

7.3 Theoretical Contributions

The main theoretical contribution of the dissertation is the development of a grey-zone sovereignty framework linking sovereignty ambiguity to market behaviour. The framework can be summarised as follows: Grey-Zone Sovereignty → Recurring Shocks → Adaptation → Interpretation → Market Reaction.

This framework begins from the claim that Taiwan's ambiguous sovereignty status is not simply a legal or diplomatic condition. It is also a recurring source of politically significant events. Diplomatic visits, military exercises, export controls, investment announcements, and sovereignty-related political signals become market-relevant because they occur within an unresolved sovereignty environment. Grey-zone sovereignty therefore operates as the upstream condition that generates repeated market-relevant shocks.

The framework also clarifies that geopolitical risk remains important. The dissertation does not reject geopolitical-risk theory. Risk events are negative on average, and some of the most negative observations are associated with military escalation, coercive economic measures, or crisis dynamics. The contribution is more specific: risk does not translate into market outcomes in a direct or uniform way. It is mediated through investor interpretation.

Adaptation is treated as an inferred mechanism. The dissertation does not observe investor beliefs directly, and it does not claim to prove that investors learned to discount Taiwan-related risk. Instead, it argues that repeated shocks may become less novel, and that this reduced novelty may influence how investors interpret later events. Adaptation is therefore a plausible explanation for some observed patterns, but it remains subject to rival explanations.

The framework's value lies in its conditionality. It explains why Taiwan-related events can produce different financial responses even when they arise from the same structural condition. A military event may be interpreted as a threat, but a repeated military event may be interpreted as familiar signalling. A semiconductor event may be interpreted as an opportunity, but an export-control event may be interpreted as a constraint. The framework therefore links grey-zone sovereignty to market behaviour without reducing either sovereignty or markets to a single causal pathway.

7.4 Implications for International Relations and International Political Economy

For International Relations, the dissertation contributes to the study of sovereignty ambiguity, contested statehood, and grey-zone competition. Taiwan is a useful case because it demonstrates how contested sovereignty generates repeated political events without necessarily producing a formal crisis each time. Taiwan's status is persistently unresolved, and this unresolved status creates recurring opportunities for signalling, pressure, recognition disputes, and military demonstrations. The dissertation shows that these events do not remain confined to diplomacy or security studies; they also become visible in financial-market behaviour.

The IR implication is therefore that sovereignty ambiguity has market consequences. Contested statehood does not only matter because it affects legal status or military risk. It also shapes how political events are interpreted by economic actors. Grey-zone competition generates

repeated signals, and those signals may become more or less market-damaging depending on how familiar, anticipated, or strategically embedded they appear.

For International Political Economy, the dissertation links geopolitical risk, financial markets, semiconductor politics, and economic security. Taiwan is useful because it is simultaneously vulnerable and strategically valuable. Its contested status creates risk, while its semiconductor centrality creates economic importance. This duality makes Taiwan a strong case for examining how markets respond when security risk and strategic economic value coexist.

The IPE implication is that markets should not be assumed to price geopolitical risk only as danger. In the Taiwan issue, some events appear to highlight policy support, strategic industry value, and supply-chain protection. At the same time, strategic importance is not automatically positive. The BIS export-control case shows that semiconductor geopolitics can also be associated with restriction and uncertainty. The dissertation therefore suggests that market responses to geopolitical shocks should be analysed as interpretive and conditional, especially in sectors tied to economic security.

7.5 Limitations

The dissertation has several limitations. The first is a small sample size. The abnormal-return analysis contains a limited number of events, and the strategic-importance category is especially small. This means that averages are sensitive to individual observations and should not be treated as stable estimates.

The second limitation concerns event classification. The distinction between geopolitical-risk events and strategic-importance events is theoretically useful, but it simplifies complex events. Some events contain both risk and strategic-importance elements. Classification choices therefore shape the empirical patterns and should be treated as transparent but contestable.

The third limitation is that adaptation is inferred rather than observed. The dissertation does not include investor surveys, analyst-expectation data, or direct evidence of investor beliefs. It observes market reactions and evaluates whether they are compatible with adaptation. This is a necessary but limited form of inference.

The fourth limitation is benchmark dependence. The abnormal returns depend on the selected benchmarks, with TWSE adjusted against NASDAQ and TSMC adjusted against SOX. These are defensible choices, but alternative benchmarks could change the magnitude or even the direction of some abnormal returns.

The fifth limitation is anticipation. Some Taiwan-related events may have been expected before the event date. If investors priced an event before Day 0, the event-window response may understate or misrepresent the market reaction. This is especially relevant for repeated military exercises and politically anticipated responses.

The sixth limitation concerns broader market conditions. The sample period overlaps with AI-sector optimism and semiconductor-cycle dynamics. Positive TSMC-related returns may partly reflect global technology-market conditions rather than Taiwan-specific geopolitical

interpretation. These limitations are not fatal flaws, but they mean that the findings should be interpreted cautiously.

7.6 Future Research

Future research should first expand the event sample. A larger dataset would allow stronger comparison across event categories, more systematic analysis of repeated shocks, and more stable estimates of average effects. It would also make it possible to separate military demonstrations, diplomatic shocks, economic coercion, and technology-policy events more precisely.

Second, future work should pursue cross-national comparison. Taiwan is a critical case, but the framework could be examined in other contexts where contested sovereignty, geopolitical pressure, and economic importance intersect. Cases involving other politically ambiguous or strategically contested territories may help assess whether grey-zone sovereignty produces similar market dynamics elsewhere.

Third, future research should incorporate more direct evidence of investor expectations. Investor surveys, analyst reports, media-event coding, or measures of news anticipation could help distinguish adaptation from anticipation. This would strengthen the empirical basis for claims about investor interpretation.

Fourth, future work should test alternative benchmarks and abnormal-return models. Comparing results across regional indices, global technology benchmarks, semiconductor peers, and market-model specifications would help assess whether the current findings are robust to benchmark choice.

Finally, future research could extend the asset coverage beyond equities. Exchange rates, credit default swaps, options-implied volatility, and sector-level indices may reveal different dimensions of Taiwan-related risk. Such extensions would help determine whether adaptation-like patterns appear across markets or are specific to equity responses.

7.7 Final Conclusion

This dissertation does not argue that geopolitical risk no longer matters. On the contrary, risk events are negative on average, and several Taiwan-related shocks are associated with substantial market weakness. The central claim is more cautious: under conditions of grey-zone sovereignty, the relationship between geopolitical risk and market outcomes is conditional rather than automatic.

Taiwan's ambiguous sovereignty status generates recurring shocks. These shocks are interpreted by investors in relation to threat, strategic importance, repetition, and surprise. Repeated shocks may become less damaging when investors interpret them as familiar, anticipated, or strategically embedded events. Yet this remains an interpretation of the observed evidence, not a proven causal mechanism.

The dissertation's contribution is therefore to show that Taiwan-related market reactions cannot be understood through geopolitical risk alone. Grey-zone sovereignty creates

vulnerability, but it also creates strategic importance. Markets appear to respond to both. The task for future research is to test more rigorously how these competing interpretations shape market behaviour. For now, the evidence supports a cautious conclusion: Taiwan-related geopolitical shocks matter for financial markets, but their effects depend on how those shocks are interpreted within a recurring grey-zone sovereignty environment.

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